Dimensional regularisation of $\mathcal{N} = 4$ super Yang–Mills theory inside the BV formalism

with the superconformal algebra $\mathfrak{psl}(4|4)$ carried by constant ghosts:
schemes, quantum action principle, one-loop components and the algebra

N4_BV_DR / theory track

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1 Conventions, names and the inherited classical data

The four-dimensional classical BV data, the equivariant action $S + S^{(1)} + S^{(2)}$ and the definitions of the obstruction, the local representative and the components are those of the heat-kernel manuscript [HK]. This note replaces the heat-kernel regulator by dimensional regularisation formulated as a BV theory in D dimensions and derives the same objects from the regulated master equation and the

quantum action principle. Numbers of [HK] are never used; the two tracks are compared only through scheme-independent data (spectra and cohomology classes, Section 5).

External input 1.1 (The four-dimensional classical data, [HK, ch. 1–2])**.** The spaces $V_{10} = V \oplus W^{\perp}_{\mathbb{R}}$, the Lie algebra $\mathfrak{g}$ with its pairing $(-,-)$, the semi-spin representations Σ, Σ^* with Clifford action ρ, pairing $(-,-)\colon \Sigma \otimes \Sigma^* \to \mathbb{C}$, Γ-pairing $(\rho(v)\zeta_1, \zeta_2) = (v, \Gamma(\zeta_1,\zeta_2))$ and antisymmetrisation q [HK, def:c1-spaces]; the fields (c, A, λ) and antifields (c^*, A^*, λ^*) with degrees and parities [HK, def:c1-fields]; the odd pairing $\langle\,,\rangle$ [HK, def:c1-pairing]; the kernel K_0, the bracket $\{f,g\} = \Delta_0(fg) - \Delta_0(f)g - (-1)^{|f|}f\Delta_0(g)$ with $\Delta_0 = -\partial_{K_0}$ [HK, def:c1-bracket]; the action

$$S = \int \mathrm{dvol}\Big(-\tfrac{1}{4}(F_A, F_A) + \tfrac{1}{2}(\lambda, \not{d}_A\lambda) + (d_A c, A^*) + ([c,\lambda],\lambda^*) + \tfrac{1}{2}([c,c],c^*)\Big) \tag{1.1}$$

[HK, eq:c1-action]; the algebra $\mathfrak{psl}(4|4)$ realised by conformal Killing vectors and spinors [HK, def:c2-even, def:c2-odd, def:c2-bracket]; the cochains, d_{CE} and the cochain bracket [HK, def:c2-cochains, def:c2-dce]; $S^{(1)}$, $S^{(2)}$ [HK, def:c2-S1, def:c2-S2] and the master equation

$$d_{CE}S_{\mathfrak{psl}(4|4)} + \tfrac{1}{2}\{S_{\mathfrak{psl}(4|4)}, S_{\mathfrak{psl}(4|4)}\} = 0, \qquad S_{\mathfrak{psl}(4|4)} = S + S^{(1)} + S^{(2)} \tag{1.2}$$

[HK, thm:c2-master]; the two-component realisation [HK, def:c8-twocomp] with $\sigma^m = (\vec{\sigma}, i)$, $\bar{\sigma}^m = (-\vec{\sigma}, i)$, $\sigma^m\bar{\sigma}^n + \sigma^n\bar{\sigma}^m = -2\delta^{mn}$, $\varepsilon^{12} = \varepsilon^{\dot1\dot2} = 1$, $(\xi,\eta) = \xi_\alpha{}^i\eta^\alpha{}_i - \bar{\xi}_{\dot\alpha i}\bar{\eta}^{\dot\alpha i}$, S_+ undotted (Srednicki's $(2,1)$), W with upper index i; for $\mathfrak{g} = \mathfrak{su}(N)$, $(X,Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY)$, $A_m = -igA^a_m T^a$, $A_I = -igA^a_I T^a$, $\lambda = -\frac{ig}{\sqrt{2}}\lambda^a T^a$, $\operatorname{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ [HK, prop:c8-twocomp]. On the Dirac column $\rho(v) = -i\gamma^\mu_S v_\mu = \gamma^\mu_W v_\mu$ gives the Weinberg form of every two-component formula [HK, prop:c8-twocomp].

Remark 1.2 (Names used in this note)**.**

- D is the regularised dimension, as in [S05]; the exterior derivative keeps the name d of [HK], and ε is not used. Laurent expansions are in $4-D$; $[f]_{(4-D)^k}$ is the coefficient of $(4-D)^k$.
- Spinor parameters are $\xi \in \Sigma$, $\eta \in \Sigma^*$ [HK, ch. 8, Names]; $Q_1, S_1, D_1, K_1, P_1, M_1, R_1$ are one-loop components (Definition 3.17).
- The translation along $v_1 \in V$ is P_{v_1} and the special conformal field $x \mapsto (x,x)v_1 - 2(v_1,x)x$ is K_{v_1} ([HK, def:c2-even] writes v_1, [HK, ch. 8] writes b); b denotes here the Nakanishi–Lautrup field of the non-minimal pair $(\bar{c}, b)$, and ξ_{GF} the gauge parameter.
- The unit is $\frac{\hbar\operatorname{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, equal to $-\frac{g^2N}{32\pi^2}$ for $\mathfrak{su}(N)$ [HK, ch. 8].
- For a vector field v, $\sigma_v := \frac{1}{D}\operatorname{div} v$ is its conformal factor in D dimensions: $\sigma_{P_{v_1}} = \sigma_{M_\omega} = 0$, $\sigma_\Delta = 2$, $\sigma_{K_{v_1}} = -2(v_1,x)$.

2 The regulated BV theories in D dimensions

A dimensional scheme is a BV theory in D dimensions: a field space, an odd pairing, a bracket and a regulated equivariant action whose failure to satisfy (1.2) is a local cochain that vanishes in the four-dimensional limit. The Clifford algebra in D dimensions exists only abstractly, and the two schemes differ in how much of the four- and ten-dimensional spinor structure survives.

2.1 Formal spaces and the four-dimensional limit

External input 2.1 (Formal spaces, [S05, §2, App. A], [G17, (2.3)], [BM77])**.** There are infinite-dimensional real inner-product spaces with formal dimensions

$$4S \subset QDS \subset Q4S, \qquad QDS = 4S \oplus Q(D{-}4)S, \qquad Q4S = QDS \oplus Q(4{-}D)S,$$

with $4S$ strictly four-dimensional, such that the orthogonal projections onto $Q4S$, QDS, $4S$, $Q(D{-}4)S$, $Q(4{-}D)S$ have traces 4, D, 4, $D-4$, $4-D$, and a translation-invariant integral $\int \frac{d^Dp}{(2\pi)^D}$ on QDS with $\int d^Dp\, f(sp) = s^{-D}\int d^Dp\, f(p)$ and $\int \frac{d^Dp}{(2\pi)^D}e^{-t|p|^2} = (4\pi t)^{-D/2}$. Integrals of polynomials vanish. Only the inner product and these traces may be used: in $Q4S$ five vectors need not be linearly dependent [S05, §2].

Definition 2.2 (Evanescent quantities and LIM)**.** A D-dependent expression is in normal form when all Clifford products are written in the antisymmetrised basis $\rho\circ q$, all contractions are carried out with the traces of Axiom 2.1, and no identity of a finite-dimensional space other than $4S$ and $W_{\mathbb{R}}^{\perp}$ is used. For an expression whose Laurent expansion in $4-D$ has no pole, LIM takes the coefficient of $(4-D)^0$ of its normal form and then applies the strictly four-dimensional (and, in Scheme D, ten-dimensional) identities: components along $Q(D{-}4)S$ and $Q(4{-}D)S$ are set to 0, Fierz identities and [ESW, Assumption 3.1] become available [BIMS20, §2]. A local functional E is evanescent when $\operatorname{LIM} E = 0$.

2.2 Scheme D: dimensional reduction from ten dimensions

Definition 2.3 (Scheme D)**.**

- Spaces: $V_{10} \mapsto V_{10}^{\mathrm{D}} := Q4S \oplus W_{\mathbb{R}}^{\perp}$ (complexified), of formal dimension 10; positions $x \in QDS$; every field is constant along $Q(4{-}D)S \oplus W_{\mathbb{R}}^{\perp}$; d acts along QDS; dvol_D is the measure of QDS.
- Fields: $A \in \Omega^0(QDS; (V_{10}^{\mathrm{D}})^\vee \otimes \mathfrak{g})$, with $A|_{QDS}$ the connection, $A|_{Q(4-D)S}$ the $4{-}D$ ϵ-scalars and $A|_{W_{\mathbb{R}}^{\perp}}$ the six scalars: a D-dimensional gauge field and $10-D$ scalars; $\lambda \in \Omega^0(QDS; \Pi\Sigma_D \otimes \mathfrak{g})$; c, antifields and degrees as in [HK, def:c1-fields].
- Clifford data: Cl_D is the algebra generated by $\rho(v)$, $v \in V_{10}^{\mathrm{D}}$, with the single relation $\rho(v_1)\rho(v_2) + \rho(v_2)\rho(v_1) = 2(v_1, v_2)$; $\Sigma_D \oplus \Sigma_D^*$ is a $\mathbb{Z}/2$-graded module on which every $\rho(v)$ is odd (a totally anticommuting chirality, [S05, (2.6)]), with an even pairing $(-,-)\colon \Sigma_D \otimes \Sigma_D^* \to \mathbb{C}$, $(\rho(v)\zeta_1, \zeta_2) = (\zeta_1, \rho(v)\zeta_2)$, and Γ defined by $(\rho(v)\zeta_1, \zeta_2) = (v, \Gamma(\zeta_1, \zeta_2))$. Traces of even products are fixed by $\mathrm{tr}_{\Sigma_D} 1 = 16$, cyclicity and the Clifford relation; the chirality-odd part of every trace is 0 [S05, footnote 1].
- Action: S_D is (1.1) with $\mathrm{dvol} \mapsto \mathrm{dvol}_D$, $F_A = dA + \frac{1}{2}[A \wedge A]$, $\not{d}_A = \rho \circ (d + [A, -])$ with all components of A, and the Hodge pairing of $\wedge^2 (V_{10}^{\mathrm{D}})^\vee$.
- Ghost spinors: Σ, Σ^* of $\mathfrak{psl}(4|4)$ embed in Σ_D, Σ_D^* as $\mathrm{Cl}(4S \oplus W_{\mathbb{R}}^{\perp})$-modules with $\Gamma(\zeta_1, \zeta_2) \in 4S \oplus W_{\mathbb{R}}^{\perp}$ for ζ_1, ζ_2 in the image.

Remark 2.4 (Two-component form of Scheme D)**.** Two-component formulas keep the form of [HK, def:c8-twocomp] with $m \in Q4S$ formal: $\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu = -2g^{\mu\nu}$, $g^\mu{}_\mu = 4$, and no further identity. In particular the completeness relation $\sigma^m_{\alpha\dot\alpha}\bar\sigma^{m\dot\beta\beta} = -2\delta_\alpha{}^\beta\delta^{\dot\beta}{}_{\dot\alpha}$, the Schouten identity and the duality $\sigma^{[m}\bar\sigma^n\sigma^{p]} \propto \epsilon^{mnpq}\sigma^q$ are unavailable: they are four-dimensional Fierz identities. The R-symmetry structure is explicit: Σ^I_{ij}, $\sum_I \Sigma^I_{ij}\Sigma^I_{kl} = 2\varepsilon_{ijkl}$, $(\rho(w)\xi)^\alpha{}_j = \varepsilon^{\alpha\beta}w_{ji}\xi_\beta{}^i$ as in [HK, def:c8-twocomp]. The kinetic terms read

$$S_D \supset \int d^Dx \Big\{ -\tfrac{1}{8}(F_{\mu\nu}, F_{\mu\nu}) - \tfrac{1}{16}(D_mA^{ij}, D_mA_{ij}) + i(\bar\lambda_{\dot\alpha i}, \bar\sigma^{m\dot\alpha\alpha}D_m\lambda_\alpha{}^i) + (D_mc, A^*_m) \Big\},$$

$\mu, \nu \in Q4S$, $m \in QDS$, with $F_{m\tilde n} = D_mA_{\tilde n}$ and $F_{\tilde m\tilde n} = [A_{\tilde m}, A_{\tilde n}]$ for $\tilde m, \tilde n \in Q(4{-}D)S$; for $\mathfrak{su}(N)$, $S_D/\hbar \supset \int d^Dx\{-\frac{1}{4}F^a_{\mu\nu}F^a_{\mu\nu} - \frac{1}{2}(D_mA_I)^a(D_mA_I)^a + i\bar\lambda^a\bar\sigma^mD_m\lambda^a\}$ [HK, prop:c8-twocomp].

2.3 Scheme B: Breitenlohner–Maison/'t Hooft–Veltman with an explicit evanescent split

Proposition 2.5 (A constraint on BMHV kinetic operators)**.** Let $V_{10} \mapsto QDS \oplus W_{\mathbb{R}}^{\perp}$ with the four-dimensional chirality $\rho(e_1e_2e_3e_4)$ on $4S$ as in [HK, prop:c1-chirality], so that $\rho(\hat v)$ commutes with the ten-dimensional chirality for $\hat v \in Q(D{-}4)S$ and maps $\Sigma \to \Sigma$. A quadratic form $\frac{1}{2}\int(\lambda, K\lambda)$ on $\Pi\Sigma \otimes \mathfrak{g}$ alone whose evanescent part is first order in $d|_{Q(D-4)S}$ and preserves the four-dimensional chirality requires an invariant linear map $J\colon W \to W^*$ of the R-symmetry group it preserves. No such map is $SU(4)$-invariant.

Proof. The pairing of [HK, def:c1-spaces] pairs Σ with Σ^*. An evanescent term $(\lambda, \rho(\hat v)\lambda)$ pairs Σ with $\rho(\hat v)\Sigma \subset \Sigma$, which needs a map $\Sigma \to \Sigma^*$. Preserving the four-dimensional chirality, it maps $S_+ \otimes W \to S_+ \otimes W^*$ and $S_- \otimes W^* \to S_- \otimes W$, so it is $1_S \otimes J$ with $J\colon W \to W^*$ intertwining the R-symmetry. W is the fundamental representation of $SU(4)$ and W^* the antifundamental;

they are inequivalent, so $J = 0$ for $SU(4)$. Adding a second spinor field in $\Pi\Sigma^* \otimes \mathfrak{g}$ with a gauge-covariant four-dimensional kinetic term changes the four-dimensional field content. □

Definition 2.6 (Scheme B)**.** Fix the real structure $J = \delta\colon W \to W^*$, $J(z)_i = \delta_{ij} z^j$, preserved by $SO(4)_R \subset SU(4)_R$.

- Spaces: $V_{10} \mapsto V_{10}^{\mathrm{B}} := QDS \oplus W_{\mathbb{R}}^{\perp}$, $QDS = 4S \oplus Q(D{-}4)S$; positions in QDS; fields constant along $W_{\mathbb{R}}^{\perp}$.
- Fields: $A \in \Omega^0(QDS; (V_{10}^{\mathrm{B}})^\vee \otimes \mathfrak{g})$, i.e. a D-dimensional gauge field $A|_{QDS} = A|_{4S} + A|_{Q(D-4)S}$ and six scalars $A|_{W_{\mathbb{R}}^{\perp}}$ (no ϵ-scalars); $\lambda \in \Omega^0(QDS; \Pi(\Sigma \otimes M) \otimes \mathfrak{g})$, where Σ is the four-dimensional semi-spin space of [HK] and M is a $\mathbb{Z}/2$-graded module of $\mathrm{Cl}(Q(D{-}4)S)$ with generators $\tau(\hat{v})$, $\tau(\hat{v}_1)\tau(\hat{v}_2) + \tau(\hat{v}_2)\tau(\hat{v}_1) = -2(\hat{v}_1, \hat{v}_2)$, an odd pairing $(m, \tau(\hat{v})m') = -(m', \tau(\hat{v})m)$ and $\mathrm{tr}_M\, 1 = 1$.
- Clifford data: $\rho(v)$ for $v \in 4S \oplus W_{\mathbb{R}}^{\perp}$ is the four-dimensional one of [HK] tensored with 1_M; for $\hat{v} \in Q(D{-}4)S$, $\rho_J(\hat{v}) := i\,\gamma_5\,(J \otimes \tau(\hat{v}))$ on $\Sigma \oplus \Sigma^*$, with J on the factor W and J^{-1} on W^*, and γ_5 the four-dimensional chirality below ($+1$ on S_+, -1 on S_-). Then $\rho(v)$, $\rho_J(\hat{v})$ satisfy the Clifford relations of QDS among themselves; with $W_{\pm}^{\perp}$ the eigenspaces ± 1 of $w_{ij} \mapsto \delta_{ik}\delta_{jl} w^{kl}$ (two $SO(4)_R$-triples), $\rho_J(\hat{v})$ anticommutes with $\rho(w)$ for w in one of them and commutes with $\rho(w)$ for w in the other. $\gamma_5 := \rho(e_1 e_2 e_3 e_4)$ is the four-dimensional chirality; it anticommutes with $\rho(4S)$ and commutes with $\rho_J(Q(D{-}4)S)$ [BM77], [BIMS20, §2].
- Action: $S_D^{\mathrm{B}} = S_{4S} + S_{\mathrm{evan}}$, where S_{4S} is (1.1) with dvol_D, all contractions over $4S \oplus W_{\mathbb{R}}^{\perp}$, the gauge kinetic term over all of $QDS \oplus W_{\mathbb{R}}^{\perp}$ on the components without $Q(D{-}4)S$-index, and

$$S_{\mathrm{evan}} = \int \mathrm{dvol}_D \Big(-\tfrac{1}{4}(F_A, F_A)\big|_{\mathrm{hat}} + \tfrac{1}{2}(\lambda, \rho_J(d_A|_{Q(D-4)S})\lambda) \Big),$$

  $(F_A, F_A)|_{\mathrm{hat}}$ being the terms of (F_A, F_A) with at least one index along $Q(D{-}4)S$. The fermion kinetic operator is $\rho(d|_{4S}) + \rho_J(d|_{Q(D-4)S})$, with inverse $\frac{\rho(p|_{4S}) + \rho_J(p|_{Q(D-4)S})}{|p|^2}$ up to the factor i and $(-,-)^{-1}$.

Remark 2.7 (Two-component and Majorana form of Scheme B)**.** J identifies $\bar{\lambda}^{\dot{\alpha}A} := \varepsilon^{\dot{\alpha}\dot{\beta}}\delta^{AB}\bar{\lambda}_{\dot{\beta}B}$, and $\Psi^A = (\lambda_\alpha{}^A, \bar{\lambda}^{\dot{\alpha}A})$, $A = 1, \dots, 4$, are four Majorana spinors. With the 't Hooft–Veltman matrices $\gamma^\mu = (\gamma^m, \hat{\gamma}^{\hat{m}})$, $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ on QDS, γ_5 built from $4S$, the kinetic term is $\frac{i}{2}\bar{\Psi}^A \gamma^\mu D_\mu \Psi^A$ in Srednicki's conventions with the Wick map of [HK, def:c8-twocomp]; its evanescent part $\frac{i}{2}\bar{\Psi}^A \hat{\gamma}^{\hat{m}} D_{\hat{m}} \Psi^A$ reads in two-component form

$$\tfrac{1}{2}\int d^D x \Big\{ \delta_{ij}\varepsilon^{\alpha\beta}(\lambda_\alpha{}^i, \tau^{\hat{m}} D_{\hat{m}} \lambda_\beta{}^j) - \delta^{ij}\varepsilon^{\dot{\alpha}\dot{\beta}}(\bar{\lambda}_{\dot{\alpha}i}, \tau^{\hat{m}} D_{\hat{m}} \bar{\lambda}_{\dot{\beta}j}) \Big\},$$

with $\tau^{\hat{m}}$ acting on the formal factor M; the four-dimensional part is $i(\bar{\lambda}_{\dot{\alpha}i}, \bar{\sigma}^{m\dot{\alpha}\alpha} D_m \lambda_\alpha{}^i)$, $m \in 4S$. The six scalars split under $SO(4)_R \cong SU(2) \times SU(2)$ into the triples $W_{\pm}^{\perp}$; the Yukawa couplings are those of [HK] on $4S \oplus W_{\mathbb{R}}^{\perp}$, equivalently $\bar{\Psi}^A(\alpha_{AB}^I + i\gamma_5 \beta_{AB}^I)[A_I, \Psi^B]$ with the 't Hooft symbols of [HK, def:c8-twocomp]. All σ-matrices in Scheme B are explicit $4S$ matrices; four-dimensional Fierz identities hold on $4S$-structures, and the formal objects are $\tau^{\hat{m}}$, $\hat{p}$, $Q(D{-}4)S$-components of A.

2.4 The non-minimal sector and gauge fixing

Definition 2.8 (Non-minimal pair)**.** Add the fields $\bar{c} \in \Omega^0(QDS; \mathfrak{g}[-1])$ and $b \in \Omega^0(QDS; \mathfrak{g})$ and their antifields $\bar{c}^*$, b^*; the coordinate functionals have the degrees

$$\bar{c}:\ -1, \qquad b:\ 0, \qquad \bar{c}^*:\ 0, \qquad b^*:\ -1,$$

total parities $\bar{c}$ odd, b even, $\bar{c}^*$ even, b^* odd; the even members of the two new pairs are $\bar{c}^*$ and b. The pairing, K_0 and the bracket extend by the rules of [HK, def:c1-pairing, def:c1-bracket] with the fibre pairing $(-,-)$ of $\mathfrak{g}$. The non-minimal action is

$$S_{\mathrm{nm}} := S_D + \int \mathrm{dvol}_D\,(b, \bar{c}^*),$$

and Q extends to the unique operator with $\frac{1}{2}\langle e, Qe\rangle = \frac{1}{2}\langle e, Qe\rangle_{\min} + \int \mathrm{dvol}_D(b, \bar{c}^*)$ preserving the pairing; it maps the slot of b to that of $\bar{c}$ and the slot of $\bar{c}^*$ to that of b^*, and $Q^2 = 0$.

Definition 2.9 (Gauge fermion and gauge-fixed action)**.** For $\xi_{\mathrm{GF}} \in \mathbb{R}$,

$$\Psi := \int \mathrm{dvol}_D(\bar{c},\ -d^*A|_{QDS} + \xi_{\mathrm{GF}}\, b), \qquad -d^*A|_{QDS} = \partial \cdot A|_{QDS},$$

of degree -1. The gauge-fixed equivariant action is $S_{\Psi,\mathfrak{psl}} := e^{\{\Psi,-\}} S_{\mathrm{nm},\mathfrak{psl}}$ (Definition 2.12); since Ψ contains no antifield, $\{\Psi,-\}$ lowers the antifield number by one and the series terminates; on functionals it is the substitution $\phi^* \mapsto \phi^* + \delta\Psi/\delta\phi$ of [HK]'s pairing conventions. Eliminating b leaves $-\frac{1}{4\xi_{\mathrm{GF}}}\int \mathrm{dvol}_D(d^*A|_{QDS}, d^*A|_{QDS})$, which with the gauge kinetic term $-\frac{1}{4}\int (dA, dA)$ gives $-\frac{1}{4}\int (A, (d^*d + \xi_{\mathrm{GF}}^{-1} dd^*)A)$: $\xi_{\mathrm{GF}} = 1$ is the Feynman gauge, $\langle A_m A_n\rangle = 2(-,-)^{-1}\delta_{mn}/|p|^2$, and in $\mathfrak{su}(N)$ components the gauge term is $-\frac{\hbar}{2\xi_{\mathrm{GF}}}(\partial \cdot A^a)^2$.

Lemma 2.10 (Gauge fixing is a canonical transformation)**.** $\{\Psi,-\}$ is an even derivation of the bracket of degree 0 commuting with d_{CE}. Hence $e^{\{\Psi,-\}}$ preserves the bracket and

$$d_{CE} S_{\Psi,\mathfrak{psl}} + \tfrac{1}{2}\{S_{\Psi,\mathfrak{psl}}, S_{\Psi,\mathfrak{psl}}\} = e^{\{\Psi,-\}}\Big(d_{CE} S_{\mathrm{nm},\mathfrak{psl}} + \tfrac{1}{2}\{S_{\mathrm{nm},\mathfrak{psl}}, S_{\mathrm{nm},\mathfrak{psl}}\}\Big).$$

The gauge-fixed operator Q^{GF} of the free part maps antifield slots to field slots, $(Q^{GF})^2 = 0$, and $[Q, Q^{GF}] = H$ with $H = |p|^2$ on every slot at $\xi_{\mathrm{GF}} = 1$; the propagator is $P_D = (Q^{GF} \otimes 1)K_0/|p|^2$ and $(Q \otimes 1 + 1 \otimes Q)P_D = K_0$ as an identity of D-dimensional kernels.

Proof. Ψ is odd of degree -1; by the Jacobi identity of the odd bracket, $\{\Psi,-\}$ is a derivation of degree 0 of $\{,\}$ and of the product, even as an operator; it is $C^\bullet(\mathfrak{psl}(4|4))$-linear because Ψ has arity 0, so it commutes with d_{CE}. The exponential of an even derivation of a bracket is an automorphism; applied to the Maurer–Cartan defect it gives the displayed identity. For the free part, $e^{\{\Psi,-\}}$ substitutes $A^*|_{QDS} \mapsto A^*|_{QDS} + \delta\Psi/\delta A$, $\bar{c}^* \mapsto \bar{c}^* + \delta\Psi/\delta\bar{c}$, $b^* \mapsto b^* + \delta\Psi/\delta b$, so the field part of the free gauge-fixed action is the quadratic form of Definition 2.9 and the ghost term $\int (dc, \delta\Psi/\delta A)$, a multiple of $\int (dc, d\bar{c})$; its inverse, written as Q^{GF}/H on the antifield slots, is the propagator, and Q^{GF} reads antifields only, hence squares to zero. The kernel identity is p-by-p: $Q(p)Q^{GF}(p) + Q^{GF}(p)Q(p) = |p|^2$ on each slot, divided by $|p|^2$. □

Remark 2.11. The heat-kernel track uses the minimal sector and a Feynman-gauge Q^{GF} with $(Q^{GF})^2 \neq 0$ [HK, ch. 3]; the composite $A|_V$–c lines of [HK, ch. 9–12] are products $P \star XP\star$ with $(P\star)^2 \propto (Q^{GF})^2$. With the non-minimal pair $(P_D\star)^2 = 0$; the corresponding contributions reappear along ξ, η, K_{v_1} as the vertices of Proposition 2.13.

2.5 The regulated equivariant action

Definition 2.12 (Transcription rules and the regulated equivariant action)**.** $S_{\mathrm{nm},\mathfrak{psl}} := S_{\mathrm{nm}} + S_D^{(1)} + S_D^{(2)} \in C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}_D))$, where the ghost algebra is the four-dimensional $\mathfrak{psl}(4|4)$ itself and $S_D^{(1)}$, $S_D^{(2)}$ are [HK, eq:c2-S1even, eq:c2-S1Q, eq:c2-S1S, eq:c2-S2] transcribed by:

- T1 $\mathrm{dvol} \mapsto \mathrm{dvol}_D$, d along QDS, all pairings as in the scheme;
- T2 Killing data in QDS: P_{v_1}, $v_1 \in 4S$; M_ω, $\omega \in \mathfrak{so}(4S)$; $\Delta = 2E$, $E(x) = x \in QDS$; K_{v_1}, $v(x) = (x,x)v_1 - 2(v_1,x)x$, $x \in QDS$; Killing spinors $x \mapsto \xi + \rho(x)\eta$, $x \in QDS$; R_r, $r \in \mathfrak{so}(W_{\mathbb{R}}^\perp)$;
- T3 Weyl factors $\frac{1}{4}\operatorname{div} v \mapsto \sigma_v$, $\frac{3}{8}\operatorname{div} v \mapsto \frac{3}{2}\sigma_v$: four-dimensional weights A: 1 (all components), λ: $\frac{3}{2}$, c: 0, $\bar{c}$: 2, b: 2; the antifields then carry D minus these weights. With the internal space $E := Q(4-D)S \oplus W_{\mathbb{R}}^\perp$ in Scheme D and $E := W_{\mathbb{R}}^\perp$ in Scheme B, every occurrence of $A|_{W_{\mathbb{R}}^\perp}$ in [HK, eq:c2-S1even, eq:c2-S1S] becomes $A|_E$: $v.A = L_v(A|_{QDS}) + v(A|_E) + \sigma_v A|_E$, and the last term of $S^{(1)}(\eta)$ is $-(\rho(A|_E)\eta, \lambda^*)$;
- T4 in Scheme B every structure built from a ghost $(\Gamma(\zeta,\lambda)$, $\rho(F_A)\zeta$, $\rho(A|_{W^\perp})\eta)$ uses components along $4S \oplus W_{\mathbb{R}}^\perp$ only;

T5 the non-minimal pair: for even x with vector field v, $S_D^{(1)}(x) \supset -\int(x.\bar c, \bar c^*) + \int(x.b, b^*)$, $x.\bar c = v(\bar c) + 2\sigma_v \bar c$, $x.b = v(b) + 2\sigma_v b$; odd x act trivially on $(\bar c, b)$; for odd x_1, x_2,

$$S_D^{(2)}(x_1, x_2) \supset \kappa \int \mathrm{dvol}_D\,([x_1, x_2].\bar c,\ b^*),$$

with $\kappa \in \{\pm 1\}$ fixed by the arity-two component of the Maurer–Cartan equation restricted to $(\bar c, b)$ (Proposition 2.13 (iii)).

Proposition 2.13 (The gauge-fixing vertices)**.** Write $x.F := \{S_D^{(1)}(x), F\}$ for a functional F (for odd x the parity sign of [HK, thm:c8-L (iv)] is $+1$), so that $x.A$ is the coefficient of A^* in $S_D^{(1)}(x)$: $\xi.A = -\Gamma(\xi, \lambda)$, $\eta.A = -\Gamma(\rho(x)\eta, \lambda)$, $v.A = L_v A + \sigma_v A|_{W^\perp}$ in the conventions of [HK, def:c2-S1].

(i) $S_\Psi^{(1)}(x) = S_D^{(1)}(x) + \{\Psi, S_D^{(1)}(x)\}$ with $\{\Psi, S_D^{(1)}(x)\} = -x.\Psi$ for even x and $+x.\Psi$ for odd x; $S_\Psi^{(2)} = S_D^{(2)} + \{\Psi, S_D^{(2)}\}$.

(ii) Odd generators (λ–$\bar c$ vertices): $\xi.\Psi = -\int \mathrm{dvol}_D\big(\bar c, \partial\cdot(\xi.A)|_{QDS}\big) = -\int \mathrm{dvol}_D\big(d\bar c, \Gamma(\xi,\lambda)|_{QDS}\big)$, and the same with $\rho(x)\eta$ in place of ξ.

(iii) Even generators: with $\partial\cdot(L_v A) = (v + 2\sigma_v)\partial\cdot A + (2-D)(d\sigma_v, A)$ for conformal Killing v on QDS,

$$v.\Psi = (4-D)\int \mathrm{dvol}_D\,\sigma_v(\bar c, \partial\cdot A + \xi_{\mathrm{GF}} b) + (2-D)\int \mathrm{dvol}_D\,(\bar c, \iota_{\mathrm{grad}\,\sigma_v} A).$$

It vanishes for P_{v_1}, M_ω, R_r; for Δ it is $2(4-D)\int(\bar c, \partial\cdot A + \xi_{\mathrm{GF}} b)$; for K_{v_1} the last term is the $A|_{QDS}$–$\bar c$ vertex $2(D-2)\int(\bar c, \iota_{v_1} A)$, equal to $4\int(\bar c, \iota_{v_1} A)$ at $D = 4$.

(iv) Arity two ($\bar c\bar c$ vertex): $\{\Psi, \kappa\int([x_1,x_2].\bar c, b^*)\} = \pm\kappa\,\xi_{\mathrm{GF}}\int \mathrm{dvol}_D\big([x_1,x_2].\bar c, \bar c\big)$, nonzero for $\xi_{\mathrm{GF}} \neq 0$ and absent in the Landau gauge.

Proof. (i) Graded symmetry $\{f,g\} = (-1)^{|f||g|}\{g,f\}$: Ψ is odd, $S^{(1)}(x)$ has parity $1+|x|$ [HK, def:c2-cochains]; $S^{(1)}$ is linear and $S^{(2)}$ quadratic in antifields not paired with fields of Ψ except $A^*|_{QDS}$, $\bar c^*$, b^*, and $\{\Psi, \{\Psi, S^{(1)}\}\} = 0$ because $\{\Psi, S^{(1)}\}$ contains no antifield of A, $\bar c$, b. (ii) $\xi.\bar c = \xi.b = 0$; the odd derivation $\xi.$ passes the odd $\bar c$ with the sign -1; $\int(\bar c, \partial\cdot X) = -\int(d\bar c, X)$; the second form uses $\xi.A = -\Gamma(\xi,\lambda)$. (iii) For a conformal Killing vector, $\partial_\mu v_\nu + \partial_\nu v_\mu = 2\sigma_v\delta_{\mu\nu}$ and $\partial^2 v_\nu = (2-D)\partial_\nu\sigma_v$, which give the stated divergence identity; then

$$\begin{aligned} v.\Psi = &\int((v+2\sigma_v)\bar c, \partial\cdot A) + \int(\bar c, (v+2\sigma_v)\partial\cdot A) + (2-D)\int(\bar c, \iota_{\mathrm{grad}\,\sigma_v} A) \\ &+ \xi_{\mathrm{GF}}\Big[\int((v+2\sigma_v)\bar c, b) + \int(\bar c, (v+2\sigma_v) b)\Big], \end{aligned}$$

and $\int((v+2\sigma_v)f, g) + \int(f, (v+2\sigma_v)g) = \int(4\sigma_v - \operatorname{div} v)(f,g) = (4-D)\int\sigma_v(f,g)$. For K_{v_1}, $\mathrm{grad}\,\sigma_v = -2v_1$. (iv) $\delta\Psi/\delta b = \xi_{\mathrm{GF}}\bar c$. □

Remark 2.14 (Two-component form of the gauge-fixing vertices)**.** For $\xi = (\xi_\alpha{}^i, \bar\xi_{\dot\alpha i})$, $\Gamma(\xi,\lambda)|_V^m = i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$ [HK, def:c8-twocomp], so

$$\xi.\Psi = -i\int d^D x\Big(\partial_m\bar c,\ \bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i + \bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\Big), \qquad m \in QDS \text{ (Scheme D)},\ m \in 4S \text{ (Scheme B)},$$

and along η the same with $\epsilon_\alpha{}^i(x) = -ix_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i}(x) = ix_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ [HK, eq:c8-killing]; this is the structure $\partial^\mu\bar c(\epsilon\sigma_\mu\bar\lambda + \lambda\sigma_\mu\bar\epsilon)$ of [MPW96, (2.8)]. The K_{v_1} vertex is $2(D-2)\int d^D x\,(\bar c, v_{1,m}A_m)$. The arity-two vertex along $(\xi, \bar\xi')$, $\xi \in S_+\otimes W$, $\bar\xi' \in S_-\otimes W^*$, is $\pm\kappa\,\xi_{\mathrm{GF}}\,i\,\bar\xi'_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i\int d^D x\,(\partial_m\bar c, \bar c)$, the counterpart of $s\,b \supset -2i\epsilon\sigma^\mu\bar\epsilon\,\partial_\mu\bar c$ in [MPW96, (2.3)].

2.6 The classical breaking

Definition 2.15 (Classical breaking)**.** $\widehat\Delta := d_{CE}S_{\Psi,\mathfrak{psl}} + \frac12\{S_{\Psi,\mathfrak{psl}}, S_{\Psi,\mathfrak{psl}}\}$, a local cochain of degree 1, with components $\widehat\Delta^{(k)}(x_1,\dots,x_k)$ normalised as in [HK, lem:c2-components]; $b_D := d_{CE} + \{S_{\Psi,\mathfrak{psl}}, -\}$.

Lemma 2.16 (Bianchi identity of the breaking)**.** $d_{CE}\widehat{\Delta} + \{S_{\Psi,\mathfrak{psl}}, \widehat{\Delta}\} = 0$ and $b_D^2 = \{\widehat{\Delta}, -\}$; $\widehat{\Delta} = e^{\{\Psi,-\}}\widehat{\Delta}_{\mathrm{nm}}$ with $\widehat{\Delta}_{\mathrm{nm}}$ the breaking of $S_{\mathrm{nm},\mathfrak{psl}}$; $\mathrm{LIM}\,\widehat{\Delta} = 0$.

Proof. The first two statements are the computation that gives $s^2 = 0$ from (1.2) in [HK, thm:c8-obstruction], with the defect kept: d_{CE} is a derivation of the cochain bracket, $d_{CE}^2 = 0$, and the Jacobi identity gives $\{S, \{S, F\}\} = \frac{1}{2}\{\{S, S\}, F\}$ for S of degree 0. The third is Lemma 2.10. At $D = 4$ in the four-dimensional realisation $S_{\mathrm{nm},\mathfrak{psl}}$ solves (1.2) [HK, thm:c2-master], the non-minimal pair being a contractible pair carrying the representation of Definition 2.12; LIM is that evaluation. □

Theorem 2.17 (Classical breaking in Scheme D)**.** Let $\mathscr{L}_0$ be the density of $S_{0,D} := \int \mathrm{dvol}_D(-\frac{1}{4}(F_A, F_A) + \frac{1}{2}(\lambda, \not{d}_A\lambda))$. In Scheme D:

(i) $\widehat{\Delta}^{(0)} = 0$ (gauge BRST exact, non-minimal sector included).

(ii) For every even generator with vector field v, $\widehat{\Delta}^{(1)}(x) = (4-D)\int \mathrm{dvol}_D\, \sigma_v\, \mathscr{L}_0$: zero along P_{v_1}, M_ω, R_r; $2(4-D)S_{0,D}$ along Δ; $-2(4-D)\int \mathrm{dvol}_D(v_1, x)\mathscr{L}_0$ along K_{v_1}. With $4 - D = 2\epsilon$ these are the BV forms of $\delta_D S_\epsilon = -\frac{2\epsilon}{g^2\mu^{2\epsilon}}\int d^D x\, L$ and of the special conformal variation $\propto \frac{4\epsilon}{g^2\mu^{2\epsilon}}\int d^D x\, x^\nu L$ of [DHKS07, (25), (27)] (up to the factor 2 of $\Delta = 2E$, the sign convention of K and the factor i of the weight e^{iS} there).

(iii) Along $\xi \in \Sigma$: $\widehat{\Delta}^{(1)}(\xi) = \mathfrak{F}(\xi) := \frac{1}{2}\int \mathrm{dvol}_D(\lambda, \rho([\xi.A, \lambda])) = -\frac{1}{2}\int \mathrm{dvol}_D(\lambda, \rho([\Gamma(\xi, \lambda), \lambda]))$, cubic in λ; $\mathfrak{F}(\xi)$ vanishes in ten dimensions by [ESW, Assumption 3.1] and is Fierz-null.

(iv) Along $\eta \in \Sigma^*$, with $\mathfrak{F}_x(\zeta)$ the density of $\mathfrak{F}(\zeta)$:

$$\widehat{\Delta}^{(1)}(\eta) = \sum_\mu \int \mathrm{dvol}_D\, x^\mu\, \mathfrak{F}_x(\rho(e_\mu)\eta) + (4-D)\mathcal{N}(\eta), \qquad \mathcal{N}(\eta) := -\tfrac{1}{2}\int \mathrm{dvol}_D(\rho(F_A)\eta, \lambda),$$

$\{e_\mu\}$ an orthonormal basis of QDS; the relative normalisation of the two terms is fixed, their common sign is that of $\xi.A = -\Gamma(\xi, \lambda)$.

(v) $\widehat{\Delta}^{(2)}(x, y) = 0$ when x or y is even; for odd x, y, $\widehat{\Delta}^{(2)}(x, y)$ is linear in λ^*, Fierz-null, and vanishes in ten dimensions by [ESW, Lemma 3.6]; $\widehat{\Delta}^{(3)}$ on three odd arguments is linear in c^* and vanishes in ten dimensions by [ESW, Lemma 3.7]; $\widehat{\Delta}^{(k\geq 4)} = 0$.

Proof. (i) The proof of $\{S, S\} = 0$ in [HK, prop:c1-cme] uses gauge covariance of F_A and $\not{d}_A$, the Leibniz rule and the Jacobi identity, all valid for formal V_{10}^{D} and any D; the non-minimal pair is contractible; $e^{\{\Psi,-\}}$ preserves the bracket.
(ii) By T3 the field coordinates transform as $\delta\phi = v(\phi) + w_\phi\sigma_v\phi$ plus the spin rotation by the antisymmetric part of dv, and antifields with $D - w_\phi$. For a covariant density f of total weight w_f, $\int \mathrm{dvol}_D\, \delta f = \int (v(f) + w_f\sigma_v f) + R = (w_f - D)\int \sigma_v f + R$, where R collects the terms with $d\sigma_v$. Antifield terms have $w_f = D$. For $S_{0,D}$, $w_f = 4$: F_A is a two-form and $L_v F_A$ has no $d\sigma_v$-term; the scalar kinetic term gives $R = -\int(\partial^2\sigma_v)(A|_E, A|_E)$ with $E = Q(4-D)S \oplus W^\perp$, zero because σ_v is affine for every generator; the fermion kinetic term gives a multiple of $\int(\lambda, \rho(d\sigma_v)\lambda)$, zero because $d\sigma_v$ is constant and the bilinear form $(\zeta_1, \rho(u)\zeta_2)$ is symmetric while λ is odd. The non-minimal and gauge-fixing parts are covered by Lemma 2.16: $\widehat{\Delta} = e^{\{\Psi,-\}}\widehat{\Delta}_{\mathrm{nm}}$ and $\widehat{\Delta}^{(1)}_{\mathrm{nm}}(x)$ contains no antifield.
(iii) In [HK, prop:c1-norm] and [ESW, Lemma 3.4] the identity $\{S, S^{(1)}(\xi)\} = 0$ uses the Clifford relation and the Bianchi identity $d(dA) = 0$ in the quadratic part, gauge covariance in the terms with antifields, and [ESW, Assumption 3.1] in the single term cubic in λ, the variation of $\frac{1}{2}(\lambda, \rho([A, \lambda]))$ along $\xi.A$. Only the last is unavailable in Cl_D.
(iv) Let $\epsilon(x) = \rho(x)\eta$. With $\xi.A = -\Gamma(\epsilon, \lambda)$, $\xi.\lambda = \frac{1}{2}\rho(F_A)\epsilon$ for position-dependent ϵ, the variation of the quadratic part is, after the integrations by parts of the constant case and the identities $\rho(v)\rho(F) = \rho(v \wedge F) + \rho(\iota_v F)$, $\sum_\mu \rho(e^\mu)\rho(\partial_\mu dA) = -\rho(d^* dA)$,

$$\tfrac{1}{2}\sum_\mu \int \mathrm{dvol}_D(\rho(e^\mu)\rho(F_A)\partial_\mu\epsilon, \lambda), \qquad \partial_\mu\epsilon = \rho(e_\mu)\eta.$$

For N, P in QDS, in $QDS \times E$ and in E, $\sum_\mu \rho(e^\mu)\rho(q(e^N \wedge e^P))\rho(e_\mu) = (D-4)$, $(D-2)$, D times $\rho(q(e^N \wedge e^P))$. The term $-(\rho(A|_E)\eta, \lambda^*)$ of $S^{(1)}(\eta)$ contributes the D-independent coefficients -2 and -4 on the last two types, which cancel them at $D = 4$ [HK, prop:c2-coeff]; the remainder is $\frac{1}{2}(D-4)\int(\rho(F_A)\eta, \lambda) = (4-D)\mathcal{N}(\eta)$. The cubic term is the x-moment of (iii).
(v) Even–even components: $v \mapsto L_v + w\sigma_v$ is a representation of the conformal algebra of QDS for every weight w, because $\sigma_{[v,v']} = v(\sigma_{v'}) - v'(\sigma_v)$. Even–odd: the brackets of conformal Killing vectors with Killing spinors are D-independent (the computation of [HK, prop:c2-special] uses $d(v,-) = 4(x,-) \wedge (v_1,-)$ and $\sigma_{K_{v_1}} = -2(v_1, x)$, valid for $x \in QDS$), and every term of $S^{(1)}(x)$ has total weight $D - \frac{1}{2}\lambda_x$, λ_x the ad_Δ-eigenvalue [HK, def:c12-gens], so the weight terms are D-independent. Odd–odd and arity three: [ESW, Lemmas 3.6, 3.7] use [ESW, Assumption 3.1] for the λ^*- and c^*-terms; the remaining terms use the Clifford relation and the Jacobi identity. Arity ≥ 4: [HK, prop:c2-tower], which uses only the field content of $S^{(2)}$. □

Theorem 2.18 (Classical breaking in Scheme B)**.** In Scheme B, with $\mathscr{L}_0$ the density of $S_{4S} + S_{\mathrm{evan}}$ without antifields:

(i) $\widehat{\Delta}^{(0)} = 0$: S_D^{B} is gauge invariant, λ being in the adjoint and d_A covariant on QDS.

(ii) $\widehat{\Delta}^{(1)} = 0$ along P_{v_1}, M_ω and along R_r for r in the stabiliser $\mathfrak{so}(4)_J$ of J; along the nine remaining generators of $\mathfrak{su}(4)_R$, $\widehat{\Delta}^{(1)}(R_r) = \frac{1}{2}\int \mathrm{dvol}_D(\lambda, [\rho_J(d|_{Q(D-4)S}), r.]\lambda)$, evanescent; along Δ, K_{v_1} the formula of Theorem 2.17 (ii) with the present $\mathscr{L}_0$.

(iii) Along ξ: $\widehat{\Delta}^{(1)}(\xi) = \mathfrak{F}^{\mathrm{B}}(\xi) + \mathcal{E}(\xi)$, with $\mathfrak{F}^{\mathrm{B}}$ the cubic term of Theorem 2.17 (iii) with $4S \oplus W^\perp_{\mathbb{R}}$-contractions and $\mathcal{E}(\xi) := \{S_{\mathrm{evan}}, S_D^{(1)}(\xi)\}$, whose quadratic part

$$\mathcal{E}(\xi)|_2 = \Big\{ -\tfrac{1}{4}\int \mathrm{dvol}_D(dA, dA)\big|_{\mathrm{hat}} + \tfrac{1}{2}\int \mathrm{dvol}_D(\lambda, \rho_J(d|_{Q(D-4)S})\lambda),$$
$$- \int \mathrm{dvol}_D(\Gamma(\xi,\lambda)|_{4S\oplus W^\perp}, A^*) + \tfrac{1}{2}\int \mathrm{dvol}_D(\rho(dA|_{4S\oplus W^\perp})\xi, \lambda^*)\Big\}$$

is nonzero and evanescent: the $Q(D-4)S$-components of the gauge field have no partner among the fermion components (field counting).

(iv) Along η: by T4 the ghost structures use $\rho(x|_{4S})\eta$, and $\widehat{\Delta}^{(1)}(\eta) = \sum_{\mu\in 4S}\int \mathrm{dvol}_D\, x^\mu \mathfrak{F}^{\mathrm{B}}_x(\rho(e_\mu)\eta) + \mathcal{E}(\rho(x|_{4S})\eta)$: the four-dimensional sandwich $\sum_{\mu\in 4S}\rho(e^\mu)\rho(q(e^N \wedge e^P))\rho(e_\mu)$ has the D-independent coefficients $0, 2, 4$, and the conformal breaking sits in $\mathcal{E}$.

(v) $\widehat{\Delta}^{(2)}(x, y) = 0$ for x, y even; along (K_{v_1}, ζ) (whose bracket is realised with $x \in QDS$ on the left and $x|_{4S}$ on the right), along (R_r, ζ) with $r \notin \mathfrak{so}(4)_J$, and on odd pairs, $\widehat{\Delta}^{(2)}$ contains evanescent terms with $\rho_J(Q(D-4)S)$, $x|_{Q(D-4)S}$ and the $Q(D-4)S$-components of Γ, besides the Fierz-null terms of Theorem 2.17 (v).

Proof. (i) Gauge transformations act on λ by $[c, -]$, commuting with ρ, ρ_J and $J \otimes \tau$. (ii) Translations and rotations of $4S$ commute with γ_5, τ and J; $r \in \mathfrak{su}(4)$ commutes with $J \otimes \tau$ exactly on $\mathfrak{so}(4)_J$. The weight computation of Theorem 2.17 (ii) applies term by term, the evanescent kinetic term having weight 4. (iii) The $4S \oplus W^\perp$-parts of the quadratic terms cancel as in four dimensions; the terms of S_{evan} vary under $\xi.A$, $\xi.\lambda$ (T4) into terms with one $Q(D-4)S$-derivative and no compensating term, the transformation of $A|_{Q(D-4)S}$ being zero by T4. (iv), (v) As (iii), with Theorem 2.17 (iv)–(v); with $\epsilon = \rho(x|_{4S})\eta$, $\partial_\mu\epsilon = \rho(e_\mu)\eta$ for $\mu \in 4S$ and 0 for $\mu \in Q(D-4)S$. □

Corollary 2.19 (Which generators fail classically)**.**

- Exact in both schemes: gauge BRST, P_{v_1}, M_ω; exact in Scheme D, and on $\mathfrak{so}(4)_J$ in Scheme B: R_r.
- Broken by the dimensionful coupling, $\widehat{\Delta} = (4-D)\times$(weighted action density): Δ, K_{v_1}, and the part $(4-D)\mathcal{N}(\eta)$ along η. The coupling has mass dimension $D-4$ in the pairing: $\int \mathrm{dvol}_D(F_A, F_A)$ has weight $4-D$.
- Broken through Fierz identities: ξ, η and the odd–odd closure (arity two and three), in both schemes.

- Broken through field counting: in Scheme B along ξ, η ($D+6$ bosonic against sixteen fermionic components); in Scheme D the counting $D+(4-D)+6=10$ is exact and the free part of $\{S_D, S_D^{(1)}(\xi)\}$ vanishes identically.

Remark 2.20 (Two-component form of the Fierz breaking)**.** For $\xi \in S_+ \otimes W$, with $\Gamma := \Gamma(\xi,\lambda)$, $\Gamma_V^m = i\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i$, $\Gamma_{kl} = 2\varepsilon_{jikl}\xi_\alpha{}^j\varepsilon^{\alpha\beta}\lambda_\beta{}^i$, $\Gamma^{kl} = 2(\xi^k\varepsilon\lambda^l - \xi^l\varepsilon\lambda^k)$ [HK, def:c8-twocomp, rem:c12-twocomp],

$$\mathfrak{F}(\xi) = -\tfrac12\int d^Dx\Big\{i(\lambda_\alpha{}^i,[\Gamma_V^m,\bar\lambda_{\dot\alpha i}])\bar\sigma^{m\dot\alpha\alpha} + \varepsilon^{\alpha\beta}(\lambda_\alpha{}^i,[\Gamma_{ij},\lambda_\beta{}^j]) \\ + i(\bar\lambda_{\dot\alpha i},[\Gamma_V^m,\lambda_\alpha{}^i])\bar\sigma^{m\dot\alpha\alpha} - \varepsilon^{\dot\alpha\dot\beta}(\bar\lambda_{\dot\alpha i},[\Gamma^{ij},\bar\lambda_{\dot\beta j}])\Big\},$$

by $(\lambda,\rho(v)\lambda') = \lambda_\alpha{}^i(\rho(v)\lambda')^\alpha{}_i - \bar\lambda_{\dot\alpha i}(\rho(v)\lambda')^{\dot\alpha i}$ and [HK, eq:c8-pairing]. In four dimensions it vanishes by $\sigma^m_{\alpha\dot\alpha}\sigma^m_{\beta\dot\beta} = -2\varepsilon_{\alpha\beta}\varepsilon_{\dot\alpha\dot\beta}$ and the Schouten identity; it is the $\mathcal{N}=4$ counterpart of $\Delta_{\text{gauge}} = \frac{i}{2}gf_{abc}(\bar\epsilon\gamma^\mu\tilde g_a)(\bar{\tilde g}_b\gamma_\mu\tilde g_c)$ of [S05, (4.4b)]. In Weinberg's form replace $\rho(v)$ by $\gamma_W^\mu v_\mu$ on the Dirac column.

Remark 2.21 (Two-component forms of the other breakings)**.** With [HK, eq:c8-pairing] and $(\rho(w)\eta)_\alpha{}^k = -w^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j$, $(\rho(w)\eta)_{\dot\beta k} = -w_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j}$:

- $\mathcal{N}(\eta) = -\frac12\int d^Dx\,[\lambda_\alpha{}^i(\rho(F_A)\eta)^\alpha{}_i - \bar\lambda_{\dot\alpha i}(\rho(F_A)\eta)^{\dot\alpha i}]$ (colour pairing understood), where the part of $\rho(F_A)\eta$ along $QDS\wedge QDS$ is

  $$(\rho(F_A)\eta)^\alpha{}_i \supset \tfrac12 F_{mn}\,\eta^\beta{}_i(\sigma^m\bar\sigma^n)_\beta{}^\alpha, \qquad (\rho(F_A)\eta)^{\dot\alpha i} \supset -\tfrac12 F_{mn}(\bar\sigma^m\sigma^n)^{\dot\alpha}{}_{\dot\beta}\,\bar\eta^{\dot\beta i},$$

  the part along $QDS\wedge E$ is $\sum_m \rho(e^m)\rho(D_mA|_E)\eta$, i.e. $(\cdot)^\alpha{}_i \supset -i\,D_mA_{ij}\,\varepsilon_{\dot\alpha\dot\beta}\bar\eta^{\dot\beta j}\bar\sigma^{m\dot\alpha\alpha}$ and $(\cdot)^{\dot\alpha i} \supset i\,\bar\sigma^{m\dot\alpha\alpha}D_mA^{ij}\varepsilon_{\alpha\beta}\eta^\beta{}_j$, and the part along $E\wedge E$ is $\frac12\sum_{w\neq w'}[A_w,A_{w'}]\rho(e^w)\rho(e^{w'})\eta$ composed from the same formulas; in Scheme D the ϵ-scalar components $A_{\tilde m}$ enter through $\rho(e^{\tilde m})$ with $\tilde m \in Q(4-D)S$ formal.
- Scheme B, R-symmetry: for $r\in\mathfrak{su}(4)$ acting by $\lambda_\alpha{}^i \mapsto r^i{}_k\lambda_\alpha{}^k$, with $r_{ik} := \delta_{ij}r^j{}_k$,

  $$\widehat\Delta^{(1)}(R_r) = \tfrac12\int d^Dx\Big\{\varepsilon^{\alpha\beta}(r_{ik}+r_{ki})(\lambda_\alpha{}^i,\tau^{\hat m}D_{\hat m}\lambda_\beta{}^k) - \varepsilon^{\dot\alpha\dot\beta}(r^{ik}+r^{ki})(\bar\lambda_{\dot\alpha i},\tau^{\hat m}D_{\hat m}\bar\lambda_{\dot\beta k})\Big\},$$

  zero exactly on $\mathfrak{so}(4)_J$ (r_{ik} antisymmetric).
- Scheme B, field counting: $\mathcal{E}(\xi)|_2$ is the bracket of the evanescent quadratic terms $-\frac18(F_{\hat mN},F_{\hat mN}) - \frac{1}{16}(\partial_{\hat m}A^{ij},\partial_{\hat m}A_{ij}) + \frac12\{\delta_{ij}\varepsilon^{\alpha\beta}(\lambda_\alpha{}^i,\tau^{\hat m}\partial_{\hat m}\lambda_\beta{}^j) - \delta^{ij}\varepsilon^{\dot\alpha\dot\beta}(\bar\lambda_{\dot\alpha i},\tau^{\hat m}\partial_{\hat m}\bar\lambda_{\dot\beta j})\}$ ($\hat m\in Q(D-4)S$, $N\in 4S\oplus Q(D-4)S$) with the linear terms of $S^{(1)}(\xi)$ in the two-component form of [HK, rem:c12-twocomp]: $T_1 = -i\int(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i, A_m^*)$, $T_2 = -\frac12\int\varepsilon_{jikl}(\xi_\alpha{}^j\varepsilon^{\alpha\beta}\lambda_\beta{}^i, A^{*kl})$, $T_3 = -\frac14\int((\partial_mA_n-\partial_nA_m)\sigma^m_{\alpha\dot\alpha}\bar\sigma^{n\dot\alpha\beta}\xi_\beta{}^i, \lambda^{*\alpha}{}_i)$, $T_4 = -\frac{i}{2}\int(\varepsilon^{\alpha\beta}\xi_\beta{}^j\sigma^m_{\alpha\dot\alpha}\partial_mA_{ij}, \bar\lambda^{*\dot\alpha i})$, $m,n\in 4S$.
- In the non-minimal Feynman gauge the ghost line is $\langle c\,\bar c\rangle \propto (-,-)^{-1}|p|^{-2}$, replacing the line $\langle c\,A_m^*\rangle$ of [HK, prop:c8-twocomp]; the lines $\langle A_mA_n\rangle$, $\langle A_{ij}A_{kl}\rangle$, $\langle\lambda_\alpha{}^i\bar\lambda_{\dot\beta j}\rangle$ are those of [HK, prop:c8-twocomp] with $|p|^{-2}$ and D-dimensional p.

3 The quantum theory: BV with dimensional regularisation

In the heat-kernel track the quantum master equation fails at scale ε through the special edges $K_\varepsilon - K_0$ and $\hbar\Delta_\varepsilon$ [HK, def:c8-ob]. In dimensional regularisation the BV Laplacian vanishes on local functionals and the bracket is exact, so the whole defect is the classical evanescent breaking $\widehat\Delta$; its loop insertions produce the local one-loop obstruction through $(4-D)\times$pole terms. This section sets up the DR forms of the transfer identity, the obstruction, the restoring counterterms, the renormalised insertions and the local representative.

3.1 The BV Laplacian and the regularised quantum action principle

Lemma 3.1 (The BV Laplacian in DR)**.** For every local functional F on $\mathscr{E}_D$, $\Delta_0F = 0$.

Proof. $\Delta_0 = -\partial_{K_0}$ contracts a field with its antifield inside one monomial with the kernel $-\delta^{(D)}(x-y)$ times the inverse fibre pairing [HK, def:c1-bracket]; with derivatives of the two letters the result is $\int \mathrm{dvol}_D\,(\cdots)\,\partial_x^\alpha \partial_y^\beta \delta^{(D)}(x-y)|_{y=x} = \int \mathrm{dvol}_D (\cdots) \int \frac{d^D p}{(2\pi)^D} (ip)^\alpha (-ip)^\beta$, the integral of a polynomial, which vanishes (Axiom 2.1). □

Theorem 3.2 (Quantum master equation in DR)**.** For a local cochain J of degree 0, $(d_{CE}+Q)J + \frac{1}{2}\{J,J\} + \hbar\Delta_0 J = (d_{CE}+Q)J + \frac{1}{2}\{J,J\}$. The regularised quantum master equation is the classical Maurer–Cartan equation, and its defect for $J = S_{\Psi,\mathfrak{psl}} - S_{\mathrm{free}}$ is $\widehat{\Delta}$ of Definition 2.15.

Proof. Lemma 3.1. □

Definition 3.3 (Regularised generating functionals)**.** For a local cochain J (interaction with counterterms), $W_D(J) := \hbar \log\,(e^{\hbar\partial_{P_D}} e^{J/\hbar})$: the connected graphs with vertices J, edges P_D of Lemma 2.10 and D-dimensional loop integrals; $\Gamma_D(J)$ is its one-particle-irreducible part, a functional of fields and antifields with values in $C^\bullet(\mathfrak{psl}(4|4))$ [HK, thm:c8-L (iv) for the products]. $S_D(\Gamma) := d_{CE}\Gamma + \frac{1}{2}(\Gamma,\Gamma)$ is the regularised Slavnov–Taylor operator, $(\,,\,)$ the bracket of [HK, def:c1-bracket] on the effective fields.

Theorem 3.4 (DR transfer identity; regularised quantum action principle)**.** Graph by graph in DR,

$$(d_{CE}+Q)\,e^{\hbar\partial_{P_D}} = e^{\hbar\partial_{P_D}}\,(d_{CE}+Q+\hbar\Delta_0), \tag{3.1}$$

hence $(d_{CE}+Q)e^{W_D(J)/\hbar} = e^{\hbar\partial_{P_D}}\,(\hbar^{-1}\mathfrak{d}_D\,e^{J/\hbar})$ with the defect vertex

$$\mathfrak{d}_D := d_{CE}J + QJ + \tfrac{1}{2}\{J,J\},$$

and in one-particle-irreducible form

$$S_D(\Gamma_D(J)) = [\mathfrak{d}_D]\cdot\Gamma_D(J), \tag{3.2}$$

the generating functional of the graphs with one vertex $\mathfrak{d}_D$ [BM77], [S05, §3, (4.3)], [BIMS20, (6.5)]. For $J = I_{\Psi,\mathfrak{psl}}$ (no counterterms), $\mathfrak{d}_D = \widehat{\Delta}$.

Proof. By [HK, lem:c8-transfer], $[\partial_\phi, Q] = \partial_{(Q\otimes 1+1\otimes Q)\phi}$, and $(Q\otimes 1 + 1\otimes Q)P_D = K_0$ (Lemma 2.10) gives $Q\partial_{P_D} = \partial_{P_D}Q + \Delta_0$; summing the exponential series gives (3.1); d_{CE} commutes with ∂_{P_D}. On $e^{J/\hbar}$, $(Q+\hbar\Delta_0)e^{J/\hbar} = \hbar^{-1}(QJ + \frac{1}{2}\{J,J\} + \hbar\Delta_0 J)e^{J/\hbar}$, and $\Delta_0 J = 0$ (Lemma 3.1). The analytic content is that every manipulation is valid for D-dimensional integrals: shifts of loop momenta, the identity kinetic operator $\times P_D = \delta$ at the regularised level, and the vanishing of scaleless integrals; these are the two ingredients of the proof of the quantum action principle in [BM77]. In Scheme B the statement is [BM77]. In Scheme D it is External input 3.5. The Legendre transform at fixed antifields turns the connected form into (3.2). □

External input 3.5 (Quantum action principle in Scheme D, [S05, §3])**.** [S05, §3] verifies in the consistent quasi-four-dimensional construction the two analytic ingredients of [BM77] (kinetic operator × propagator $= \delta$ for D-dimensional momenta and γ-matrices of $Q4S$; vanishing of scaleless integrals) and states that the combinatorial part of the proof of [BM77] carries over; the combinatorics is not written out there [LIT, §2.1]. Theorem 3.4 in Scheme D rests on this input; gate G3b tests (3.2) graph by graph at one loop.

Remark 3.6 (The breaking as the BV Laplacian of DR)**.** In [Bar98, §2.8, (129)–(132)] the evanescent breaking of the regularised master equation is coupled to a global odd source ρ^*, and $(\tau/(-i\hbar))\,\partial^L/\partial\rho^*$ takes the place of the BV Laplacian, whose formal term vanishes because $\delta^{(D)}(0) = 0$. Theorem 3.4 is this statement in the conventions of [HK], with $\widehat{\Delta}$ in the role of the source term.

Remark 3.7 (Dictionary with the heat-kernel defect)**.**

heat kernel [HK]	dimensional regularisation
special edge $K_\varepsilon - K_0$, $\hbar\Delta_\varepsilon$	none: $\Delta_0 F = 0$, exact bracket
defect vertex $(d_{CE}+Q)J + \frac{1}{2}\{J,J\}_\varepsilon + \hbar\Delta_\varepsilon J$	$\mathfrak{d}_D = d_{CE}J + QJ + \frac{1}{2}\{J,J\}$; classical part $\widehat{\Delta}$, evanescent
ε^{-m}, $\log\varepsilon$	poles $(4-D)^{-1}$ only at one loop
$[\varepsilon^0]$ of cycles with one special edge	LIM of cycles with one evanescent vertex (Lemma 3.10)
bridge + counterterm partner $\to 0$	evanescent tree vertex $\times$ subtracted subgraph $\to 0$
endpoint T, $\log T$ along K	no endpoint; residues are infrared safe (Proposition 3.25)
$(Q^{GF})^2 \neq 0$, composite lines	$(Q^{GF})^2 = 0$; vertices $\{\Psi, S^{(1)}(x)\}$, $\{\Psi, S^{(2)}\}$

The correspondence is structural; numbers are computed separately in each track.

3.2 Renormalisation, the renormalised Slavnov–Taylor identity and the obstruction

Definition 3.8 (Counterterms and renormalised effective action)**.** The minimal counterterms I_D^{CT} are, order by order in $\hbar$ with lower-order counterterms inserted, the pole parts in $4-D$ of the one-particle-irreducible graphs, as local cochains in normal form; at one loop $I_D^{CT,(1)} = (4-D)^{-1}\,[\Gamma_D^{(1)}]_{(4-D)^{-1}}$. A finite counterterm is a local cochain $F = \hbar F_1 + \hbar^2 F_2 + \cdots$ of degree 0 and Weyl weight 4 (counted as in [HK, def:c8-quant]), given with a D-dimensional lift. With $J := I_{\Psi,\mathfrak{psl}} - I_D^{CT} - F$,

$$\Gamma_{\mathrm{ren}} := \mathrm{LIM}\,\Gamma_D(J), \qquad S(\Gamma) := d_{CE}\Gamma + \tfrac{1}{2}(\Gamma,\Gamma) \text{ (four-dimensional)}.$$

Theorem 3.9 (Renormalised Slavnov–Taylor identity)**.**

$$S(\Gamma_{\mathrm{ren}}) = \mathrm{LIM}\Big([\mathfrak{d}_D]\cdot\Gamma_D(J)\Big), \qquad \mathfrak{d}_D = \widehat{\Delta} - b_D(I_D^{CT}+F) + \tfrac{1}{2}\{I_D^{CT}+F, I_D^{CT}+F\}.$$

At one loop

$$S(\Gamma_{\mathrm{ren}}) = \hbar\,(\mathrm{Ob} - sF_1) + O(\hbar^2), \qquad \mathrm{Ob} := \mathrm{LIM}\Big([\widehat{\Delta}\cdot\Gamma_D]^{(1)} - b_D I_D^{CT,(1)}\Big), \tag{3.3}$$

with $s = d_{CE} + \{S_{\Psi,\mathfrak{psl}}, -\}$ at $D=4$ and $[\widehat{\Delta}\cdot\Gamma_D]^{(1)}$ the one-loop graphs with one vertex $\widehat{\Delta}$, the factor $\hbar$ removed.

Proof. Theorem 3.4 with $\mathfrak{d}_D$ expanded at $J = I_{\Psi,\mathfrak{psl}} - I_D^{CT} - F$: the order-$\hbar^0$ part is $\widehat{\Delta}$, the linear terms are $-b_D(I_D^{CT}+F)$. The left side has a limit; so does the right side, and the order-$\hbar$ part is the tree insertion of $\widehat{\Delta}$ (with $\mathrm{LIM}\,\widehat{\Delta}=0$), the one-loop graphs with one $\widehat{\Delta}$, and $-b_D I_D^{CT,(1)} - \hbar b_D F_1$. $\mathrm{LIM}(b_D F_1) = sF_1$ because F_1 is finite and $b_D \to s$; the double insertions $\widehat{\Delta}\otimes F_1$ are of order $\hbar^2$. This is [BIMS20, (6.3)–(6.6)] in BV form. □

Lemma 3.10 (Finite parts of evanescent insertions; Bonneau-type identity)**.** Let X_D be a local vertex whose normal form is polynomial in D, and write the sum of the one-loop graphs with one vertex X_D (other vertices from $I_{\Psi,\mathfrak{psl}}$) as $(4-D)^{-1}\mathcal{R}_X(D) + \mathcal{F}_X(D)$, with $\mathcal{R}_X(D)$ local and polynomial in D (Theorem 4.2) and $\mathcal{F}_X$ analytic at $D=4$. If $\mathrm{LIM}\,X_D = 0$, then $\mathrm{LIM}\,\mathcal{R}_X(4) = 0$, $\mathrm{LIM}\,\mathcal{F}_X(4) = 0$, and

$$\mathrm{LIM}\Big((4-D)^{-1}\mathcal{R}_X(D) + \mathcal{F}_X(D)\Big) = -\,\mathrm{LIM}\,\partial_D\mathcal{R}_X(D)\big|_{D=4}.$$

Proof. In normal form every structure of a graph with an evanescent vertex is either explicitly multiplied by a trace of the projections onto $Q(D{-}4)S$ or $Q(4{-}D)S$, i.e. by $\pm(D-4)$, or carries an uncontracted evanescent component, or (Scheme D) a Fierz-null combination; each is annihilated by LIM at $D=4$. Hence $\mathcal{R}_X(D) = (D{-}4)\mathcal{R}'_X + \mathcal{R}_X^{\mathrm{ev}} + O((D{-}4)^2)$ with $\mathrm{LIM}\,\mathcal{R}_X^{\mathrm{ev}} = 0$,

and the same for $\mathcal{F}_X$; $(4-D)^{-1}(D-4)\mathcal{R}'_X = -\mathcal{R}'_X$. In Scheme B with X_D proportional to the evanescent metric this is [Bon80], [BIMS20, (6.14)]; in Scheme D the Fierz-null case uses that the four- and ten-dimensional evaluation of each normal-form coefficient at $D = 4$ is a computation in the strictly finite-dimensional algebra, where the vertex vanishes. □

Theorem 3.11 (One-loop obstruction theory in DR)**.**

(i) Ob is a local cochain of degree 1. With $b_D = b_0 + (4-D)b_1 + O((4-D)^2)$ (explicit D-dependence of the regulated classical cochain),

$$\mathrm{Ob} = -\mathrm{LIM}\,\partial_D \mathcal{R}_{\widehat{\Delta}}(D)\big|_{D=4} - \mathrm{LIM}\, b_1 [\Gamma_D^{(1)}]_{(4-D)^{-1}}.$$

(ii) $s\,\mathrm{Ob} = 0$ (Wess–Zumino consistency).

(iii) $\hbar F_1$ is a one-loop quantisation, $S(\Gamma_{\mathrm{ren}}) = O(\hbar^2)$, exactly when $sF_1 = \mathrm{Ob}$.

(iv) One-loop quantisations form an affine space over the s-closed local cochains of degree 0; differences $s\phi$ are field redefinitions; the classes of $H^0(s)$ include the coupling direction $S - S^{(2)}$ and the outer direction G_B [HK, thm:c8-obstruction (iii), (iv), prop:c8-B]. An anomaly is a nonzero class $[\mathrm{Ob}] \in H^1(s)$; a change of the D-dimensional lifts of $S_D^{(k)}$, Ψ or F by evanescent local functionals changes Ob by an s-exact cochain.

Proof. (i) Laurent orders in (3.3): the pole parts of $[\widehat{\Delta}\cdot\Gamma_D]^{(1)}$ and of $b_D I_D^{CT,(1)} = (4-D)^{-1} b_D [\Gamma_D^{(1)}]_{(4-D)^{-1}}$ cancel because the left side of Theorem 3.9 is finite; the order $(4-D)^0$ of the first is Lemma 3.10 for $X_D = \widehat{\Delta}$, of the second $b_1[\Gamma_D^{(1)}]_{(4-D)^{-1}}$. Both are local. (ii) For every functional Γ of degree 0, $d_{CE}S(\Gamma) + (\Gamma, S(\Gamma)) = 0$ by $d_{CE}^2 = 0$, the derivation property of d_{CE} and the Jacobi identity; at order $\hbar$ with $S(\Gamma_{\mathrm{ren}}) = \hbar\,\mathrm{Ob}$ ($F_1 = 0$) and $d_{CE} + (\Gamma_{\mathrm{ren}}, -) = s + O(\hbar)$ this is $s\,\mathrm{Ob} = 0$. (iii) Theorem 3.9. (iv) The first statements are four-dimensional and are those of [HK, thm:c8-obstruction]. For a change E of a lift, $\widehat{\Delta} \mapsto \widehat{\Delta} + b_D E + \frac{1}{2}\{E, E\}$, and by (3.2) applied to the insertion of E, $\mathrm{LIM}[b_D E\cdot\Gamma_D]^{(1)} = s\,\mathrm{LIM}[E\cdot\Gamma_D]^{(1)} + O(\hbar)$, where $\mathrm{LIM}[E\cdot\Gamma_D]^{(1)}$ is local by Lemma 3.10; the change of I^{CT} is the pole part of the same graphs. □

Theorem 3.12 (Restoring counterterms: what vanishes in each scheme)**.**

(i) In both schemes $\mathrm{Ob}^{(0)} = 0$: the gauge Slavnov–Taylor identity holds without finite counterterm, and $F_1^{(0)}$ can be taken s-closed.

(ii) $\mathrm{Ob}^{(1)}(x) = 0$ for $x = P_{v_1}, M_\omega$ in both schemes and for $x = R_r$ in Scheme D and on $\mathfrak{so}(4)_J$ in Scheme B.

(iii) Along Δ: $\mathrm{Ob}^{(1)}(\Delta) = \mathrm{LIM}(2(4-D)[S_{0,D}\cdot\Gamma_D]^{(1)}) - \mathrm{LIM}\, b_1[\Gamma_D^{(1)}]_{(4-D)^{-1}}$, where the first term is s-exact and b_1 is the explicit $(4-D)$-term $-2\{\Psi, -\}$ of $S_\Psi^{(1)}(\Delta) = S_D^{(1)}(\Delta) - 2(4-D)\Psi$ (Proposition 2.13 (iii)). A component outside $\mathrm{Im}\, s$ along $\int(F_A, F_A)$ would be a trace-anomaly class (gate G4).

(iv) Along ξ, η, K_{v_1}, along R_r with $r \notin \mathfrak{so}(4)_J$ (Scheme B) and in arity two, Ob is given by (i) of Theorem 3.11 with the vertices of Theorems 2.17, 2.18; the restoring counterterms solve the arity components of $sF_1 = \mathrm{Ob}$, [HK, eq:c12-eqs], with all free directions.

Proof. (i) $\widehat{\Delta}^{(0)} = 0$ (Theorems 2.17 (i), 2.18 (i)) and the arity-zero part of b_D has no explicit D-dependence: the ghost, gauge-fixing and antifield terms of S_Ψ carry D-independent coefficients. Theorem 3.11 (i) gives $\mathrm{Ob}^{(0)} = 0$. (ii) $\widehat{\Delta}^{(1)}(x) = 0$, $\widehat{\Delta}^{(0)} = 0$ and the arity-one part of b_D along x is D-independent; the arity-one component of $[\widehat{\Delta}\cdot\Gamma_D]$ along x contains only $\widehat{\Delta}^{(0)}$ and $\widehat{\Delta}^{(1)}(x)$. (iii) $\widehat{\Delta}^{(1)}(\Delta) = 2(4-D)S_{0,D}$. The full action inserted into a one-loop one-particle-irreducible graph multiplies it by (vertices − propagators) = 0, so $[S_{0,D}\cdot\Gamma_D]^{(1)} = -[(S_\Psi - S_{0,D})\cdot\Gamma_D]^{(1)}$. At arity zero $S_\Psi - S_{0,D} = e^{\{\Psi,-\}}\{N, S_{\mathrm{nm}}\} = \{e^{\{\Psi,-\}}N, S_\Psi\}$ with $N = \int(c^*, c) - \int(\bar{c}^*, \bar{c})$ up to signs, because every antifield term of S_{nm} has ghost-minus-antighost count one more than its antifield count; by Lemma 3.16 the insertion of $s_D(e^{\{\Psi,-\}}N)$ is S_{Γ_D} of the insertion of $e^{\{\Psi,-\}}N$ plus a double insertion with $\widehat{\Delta}$, and $(4-D)$ times the latter has LIM = 0 by Lemma 3.10. Hence the

first term is $s(-2\,\mathrm{LIM}\,\mathcal{R}(e^{\{\Psi,-\}}N\cdot\Gamma_D))$. (iv) Definitions. □

3.3 The gauge parameter

Theorem 3.13 (Nielsen identity in DR-BV)**.** Adjoin to the cochains an even parameter ξ_{GF} of degree 0 and an odd constant χ of degree 1, with differential $d_{CE}+\chi\partial_{\xi_{\mathrm{GF}}}$. Then

$$S_{\mathrm{ext}} := S_{\Psi,\mathfrak{psl}} + \chi\,\partial_{\xi_{\mathrm{GF}}}\Psi, \qquad \partial_{\xi_{\mathrm{GF}}}\Psi = \int \mathrm{dvol}_D(\bar{c},b),$$

satisfies $(d_{CE}+\chi\partial_{\xi_{\mathrm{GF}}})S_{\mathrm{ext}} + \frac{1}{2}\{S_{\mathrm{ext}},S_{\mathrm{ext}}\} = \widehat{\Delta}$, with no χ-component. The χ-component of the regularised identity (3.2) is

$$\partial_{\xi_{\mathrm{GF}}}\Gamma_D = \pm(\Gamma_D,\Gamma_{D,\chi}) + [\widehat{\Delta}\otimes\partial_{\xi_{\mathrm{GF}}}\Psi]\cdot\Gamma_D, \qquad \Gamma_{D,\chi} := [\partial_{\xi_{\mathrm{GF}}}\Psi]\cdot\Gamma_D,$$

and after renormalisation with the χ-components of F_1 chosen by the χ-component of $sF_1 = \mathrm{Ob}$, $\partial_{\xi_{\mathrm{GF}}}\Gamma_{\mathrm{ren}} = \pm(\Gamma_{\mathrm{ren}},\Gamma_{\mathrm{ren},\chi})$: a change of ξ_{GF} is a canonical transformation of the renormalised theory. Consequently every quantity of Theorem 3.29 is independent of ξ_{GF}, while representatives such as the entries of Q_1 depend on it through law (a) of Proposition 3.27.

Proof. $\partial_{\xi_{\mathrm{GF}}}e^{\{\Psi,-\}} = \{\partial_{\xi_{\mathrm{GF}}}\Psi,-\}e^{\{\Psi,-\}}$ because $\{\Psi,\partial_{\xi_{\mathrm{GF}}}\Psi\} = 0$ (both contain fields only); so $\chi\partial_{\xi_{\mathrm{GF}}}S_{\Psi,\mathfrak{psl}} = \chi\{\partial_{\xi_{\mathrm{GF}}}\Psi, S_{\Psi,\mathfrak{psl}}\}$, which cancels the χ-linear part of $\frac{1}{2}\{S_{\mathrm{ext}},S_{\mathrm{ext}}\}$ by graded symmetry, $\chi^2=0$ and $d_{CE}\partial_{\xi_{\mathrm{GF}}}\Psi = 0$. Theorem 3.4 applies to the extended complex. At one loop the double insertion is local in the limit by Lemma 3.10, and its s-closedness follows from Theorem 3.11 (ii) in the extended complex. A canonical transformation generated by $\delta\xi_{\mathrm{GF}}\,\Gamma_{\mathrm{ren},\chi}$ conjugates the quantum differential of Definition 3.15. □

3.4 Renormalised insertions and the local representative

Definition 3.14 (Renormalised insertions)**.** For a local cochain M of degree $|M|$, a lift M_D ($\mathrm{LIM}\,M_D = M$), a scheme ψ for operators [HK, def:c8-renorm] and a square-zero parameter δ of degree $-|M|$,

$$M(D) := M_D - \partial_\delta I_D^{CT}(M) + \hbar\,\psi(M), \qquad [M]_R := \mathrm{LIM}\,\partial_\delta\Gamma_D(J+\delta M(D)),$$

$\partial_\delta I_D^{CT}(M)$ being the pole part of the graphs through the vertex δM. $\psi=0$ is minimal subtraction; MS-bar is ψ equal to $(\log 4\pi - \gamma_E)$ times the residue map at one loop.

Definition 3.15 (Quantum differential)**.** $S_\Gamma := d_{CE} + (\Gamma_{\mathrm{ren}},-)$ on functionals of the effective fields; $S_\Gamma^2 = 0$ when $S(\Gamma_{\mathrm{ren}}) = 0$.

Lemma 3.16 (Quantum action principle for insertions)**.**

$$S_{\Gamma_D}\big(\partial_\delta\Gamma_D(J+\delta M(D))\big) = [b_D M(D)]\cdot\Gamma_D + [\mathfrak{d}_D\otimes M(D)]\cdot\Gamma_D,$$

with $S_{\Gamma_D} = d_{CE} + (\Gamma_D,-)$ and the double insertion of the action defect and of $M(D)$; in the limit

$$S_\Gamma[M]_R = \mathrm{LIM}\Big([b_D M(D)]\cdot\Gamma_D + [\mathfrak{d}_D\otimes M(D)]\cdot\Gamma_D\Big).$$

Proof. The δ-linear part of (3.2) for the family $J+\delta M(D)$, whose defect vertex is $\mathfrak{d}_D + \delta\, b_D M(D)$ up to the sign of δ [HK, lem:c8-flow]. □

Definition 3.17 (Local representative and components in DR)**.** A local representative of M is a local cochain $L(M)$ of degree $|M|+1$ with

$$S_\Gamma[M]_R = [L(M)]_R. \tag{3.4}$$

For a local functional O write $L(O) = \sum_k L(O)^{(k)}$ by arity and $L = s + \hbar L_1 + \cdots$. The one-loop components are, as in [HK, def:c8-components],

$$Q_1(O) = \hbar L_1(O)^{(1)}(\xi), \quad S_1(O) = \hbar L_1(O)^{(1)}(\eta), \quad D_1(O) = \hbar L_1(O)^{(1)}(\Delta),$$
$$K_1(O) = \hbar L_1(O)^{(1)}(K_{v_1}), \quad P_1(O) = \hbar L_1(O)^{(1)}(P_{v_1}),$$

$M_1(O)$, $R_1(O)$ along $\mathfrak{so}(4S)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, the BRST correction $\hbar L_1(O)^{(0)}$ and the arity-two values $\hbar L_1(O)^{(2)}(x_1, x_2)$. Two-component: $Q_1(O) = \xi_\alpha{}^i Q_1(O)^\alpha{}_i + \bar\xi_{\dot\alpha i}\bar Q_1(O)^{\dot\alpha i}$ along Q^i_α, $\bar Q_{\dot\alpha i}$, and $S_1(O) = \eta^\alpha{}_i S_1(O)_\alpha{}^i + \bar\eta^{\dot\alpha i}\bar S_1(O)_{\dot\alpha i}$ along $S^\alpha{}_i$, $\bar S^{\dot\alpha i}$ [HK, def:c8-twocomp].

Theorem 3.18 (Well-posedness)**.** Assume $S(\Gamma_{\mathrm{ren}}) = O(\hbar^2)$.

(i) If $[N]_R = 0$ for a local cochain N then $N = 0$; $L(M)$ is unique.

(ii) At one loop $L(M)$ exists for every local M, and $L(M) = sM + \hbar L_1(M)$ with $\hbar L_1$ given by Theorem 3.20.

(iii) $L \circ L = 0$; with $\Lambda = L - d_{CE}$ this is the Maurer–Cartan equation $d_{CE}\Lambda + \Lambda d_{CE} + \Lambda\Lambda = 0$ of [HK, thm:c8-L (iv)], and $x \mapsto -\Lambda^{(1)}(x)$ is a representation of $\mathfrak{psl}(4|4)$ on the cohomology of $\Lambda^{(0)}$ [HK, cor:c8-homotopy].

Proof. (i) The tree-level part of $[N]_R$ is the tree flow of N, injective by the leg-counting induction of [HK, thm:c8-L (i)]. (ii) Theorem 3.20. (iii) $S_\Gamma^2 = 0$ and (i). □

Definition 3.19 (Operator breaking)**.** For a local functional O with lift O_D and a generator x, $\widehat\Delta_O(x) := \{S_D^{(1)}(x), O_D\} - (\{S^{(1)}(x), O\})_D$, an evanescent local functional; $\widehat\Delta_O := s_D O_D - (sO)_D$ in arity zero. For $O = \int \mathrm{dvol}_D\, f\,\Phi$ with Φ a product of letters of total weight w_Φ and f a test function, $\{S_D^{(1)}(v), O_D\} = \int \Phi\,(-v(f) + (w_\Phi - D)\sigma_v f)$ for even v, hence

$$\widehat\Delta_O(v) = (4 - D)\int \mathrm{dvol}_D\,\sigma_v\, f\,\Phi.$$

Theorem 3.20 (DR master formula at one loop)**.** Let $O = \int \omega\, X_1 \cdots X_n$ have degree 0, $\psi = 0$, and $\hbar F_1$ solve $sF_1 = \mathrm{Ob}$ in arity zero and in arity one along x. Then

$$\begin{aligned}\hbar L_1(O)^{(1)}(x) &= \sum_{a<b} \pm \Upsilon_x^D(X_a, X_b) + \sum_a \Upsilon_x^D(X_a) - \{\hbar F_1^{(1)}(x), O\},\\ \hbar L_1(O)^{(0)} &= \sum_{a<b} \pm \Upsilon_0^D(X_a, X_b) - \{\hbar F_1^{(0)}, O\},\end{aligned} \tag{3.5}$$

with the Koszul signs of [HK, eq:c12-pairsum] and the two-body kernel

$$\Upsilon_x^D(X_a, X_b) := -\hbar\,\mathrm{LIM}\,\partial_D\Big(\mathcal{R}_x^{\mathrm{cyc}}(X_a, X_b) + \mathcal{R}_x^{\mathrm{op}}(X_a, X_b)\Big)\Big|_{D=4} + \hbar\,\mathrm{LIM}\,\mathcal{C}_x(X_a, X_b),$$

where $\mathcal{R}_x^{\mathrm{cyc}}$ is the residue polynomial of the one-loop cycles through the pair with exactly one vertex from $\widehat\Delta^{(1)}(x)$ on the cycle, $\mathcal{R}_x^{\mathrm{op}}$ that of the cycles through the pair of $\widehat\Delta_{X_aX_b}(x)$, and

$$\mathcal{C}_x(X_a, X_b) := (4 - D)^{-1}\Big(\{S_D^{(1)}(x), \mathcal{R}(X_aX_b)\} - \mathcal{R}(\{S^{(1)}(x), X_aX_b\})_D\Big) - \partial_\delta I_D^{CT,(1)}(X_aX_b)(x)$$

is the commutator of the classical transformation with the counterterm map, $\mathcal{R}(M) := [\partial_\delta I_D^{CT}(M)]_{(4-D)^{-1}}$, together with the arity-one counterterm of the pair; $\Upsilon_x^D(X_a)$ is the same construction for the one-letter part of $\widehat\Delta_O(x)$, and Υ_0^D the arity-zero analogue.

Proof. Lemma 3.16 with $M = O$ at order $\hbar$, δ-linear, arity one along x. The graphs fall into the classes of [HK, thm:c12-master (i)], with the special edge replaced by an evanescent vertex: (F) one-loop graphs of the vertices $\{S_D^{(1)}(x), O_D\}$, $s_D O_D$, which with the counterterm trees of the lifts of $\{S^{(1)}(x), O\}$, sO form their renormalised insertions; (C2) cycles through two letters of O with

one vertex of $\widehat{\Delta}^{(1)}(x)$; (Cop) cycles through two letters of $\widehat{\Delta}_O(x)$; (Br) an evanescent vertex on a tree attached to a subtracted subgraph, whose limit is LIM of a finite quantity times an evanescent tree, zero; (Ac) cycles of J alone, whose limit is the tree flow of $\hbar(\mathrm{Ob}-sF_1)=0$ through arity one along x; (Ct) the counterterm vertices $-\{S_D^{(1)}(x),\partial_\delta I_D^{CT}(O)\}$, $-\partial_\delta I_D^{CT,(1)}(O)(x)$, $-\{I_D^{CT,(1)}(x)+\hbar F_1^{(1)}(x),O\}$. The poles cancel because the left side of Lemma 3.16 has a limit; the finite parts of (C2), (Cop) are Lemma 3.10; the finite part of (Ct) is $\mathcal{C}_x$; $\{I_D^{CT,(1)}(x),O\}$ is a pole term cancelling against (C2). At one loop the cycle through O uses exactly two letters and a tree uses one [HK, thm:c10-decomp], which gives the pair sum and the one-body terms. Injectivity (Theorem 3.18 (i)) identifies L. □

Corollary 3.21 (Exact components)**.** With covariant lifts O_D, $\psi=0$ and $F_1^{(0)}=F_1^{(1)}(P_{v_1})=F_1^{(1)}(M_\omega)=F_1^{(1)}(R_r)=0$:

- $\hbar L_1(O)^{(0)}=0$ on every local O in both schemes: the BRST differential on operators is uncorrected at one loop;
- $P_1=M_1=0$ in both schemes; $R_1=0$ in Scheme D, and in Scheme B on $\mathfrak{so}(4)_J$.

Proof. For these x, $\widehat{\Delta}^{(1)}(x)=0$, $\widehat{\Delta}_O(x)=0$ for covariant lifts, and the counterterm map commutes with the D-independent classical transformation, so $\mathcal{C}_x=0$; arity zero likewise with $\widehat{\Delta}^{(0)}=0$. □

3.5 The dilatation from the renormalisation group

Theorem 3.22 (D_1 in DR)**.** For O of degree 0 with covariant lift and $\psi=0$,

$$D_1(O)=-2\,\mathrm{LIM}\,\mathcal{R}(O)-\{\hbar F_1^{(1)}(\Delta),O\}+[s,\varphi_O],$$

where $\mathcal{R}(O)=[\partial_\delta I_D^{CT}(O)]_{(4-D)^{-1}}$ is the residue of the one-particle-irreducible pair kernels (Theorem 4.2), $\hbar F_1^{(1)}(\Delta)$ solves the arity-one equation along Δ (a field-counting functional plus free directions, Theorem 3.12 (iii)) and φ_O is a local functional. On the cohomology of $\Lambda^{(0)}$, in particular on gauge-invariant O, $D_1=-2\times$(the residue of the complete one-loop counterterm of O, the self-energies of its letters included), independent of ξ_{GF}. Equivalently, with $[O]_R=Z_O\,O_D$ the dilatation Ward identity of Theorem 3.4 along Δ is the Callan–Symanzik equation

$$\tfrac{1}{2}D_1=\gamma_O=-\mu\partial_\mu\log Z_O\big|_{(4-D)^0},$$

where $\mu\partial_\mu$ of the bare pairing at fixed renormalised pairing is $(D-4)$ times the coupling derivative, and $\beta=0$ at one loop in both schemes. Evanescent counterterms $(4-D)^{-1}E$, $\mathrm{LIM}\,E=0$, contribute $-2\,\mathrm{LIM}\,E=0$ at one loop; at two loops their finite renormalisation enters and changes $\gamma^{(2)}$ on the physical block by a commutator $[\gamma^{(1)},r]$ with the finite one-loop mixing r of [DG91], [HN95], so the two-loop spectrum is independent of it when $\beta=0$.

Proof. Theorem 3.20 along Δ. (Cop) By Definition 3.19, $\widehat{\Delta}_O(\Delta)=2(4-D)O_D$, whose cycles give $2(4-D)\cdot(4-D)^{-1}\mathcal{R}(O)=2\mathcal{R}(O)$. (Ct) $\{S_D^{(1)}(\Delta),\int f\mathcal{R}(\Phi)\}-\mathcal{R}(\{S^{(1)}(\Delta),O\})_D=2(4-D)\mathcal{R}(O)$, entering with the sign of $-\partial_\delta I^{CT}$: $-2\mathcal{R}(O)$. (C2) $\widehat{\Delta}^{(1)}(\Delta)=2(4-D)S_{0,D}$: the full action inserted into a one-loop graph through O multiplies it by (vertices of I − propagators) $=-1$, so $2(4-D)[S_{0,D}\otimes O]^{(1)}\to-2\mathcal{R}(O)-2\,\mathcal{R}\big((S_\Psi-S_{0,D})\otimes O\big)$; $S_\Psi-S_{0,D}$ is s-exact at arity zero (proof of Theorem 3.12 (iii)), and by Lemma 3.16 its insertion, together with the explicit vertex $-2(4-D)\Psi$ of $S_\Psi^{(1)}(\Delta)$, contributes $[s,\varphi_O]$. Total $-2\mathcal{R}(O)$, plus $-\{\hbar F_1^{(1)}(\Delta),O\}$. On cohomology $[s,\varphi_O]$ vanishes, and the self-energies of the letters enter through $F_1^{(1)}(\Delta)$; ξ_{GF}-independence is Theorem 3.13. The Callan–Symanzik form: $(4-D)\int\sigma_v\mathscr{L}_0$ is $(4-D)$ times the coupling derivative of the action at fixed fields, and the bare coupling has dimension $D-4$ in the pairing. At two loops: [DG91], [HN95]. □

Remark 3.23 (Normalisation check)**.** For the Konishi bilinear $\sum_I (A_I, A_I)$ the scheme-independent value is $\gamma = \frac{3g^2N}{4\pi^2}$, i.e. $D_1 = 2\gamma = -48$ units, and $D_1 = 0$ on the traceless bilinears (the **20′**); these are gates G5, G6 of Section 6.

3.6 Special conformal transformations from the D-dimensional conformal Ward identity

Theorem 3.24 (K_1 in DR)**.** Let O sit at x_O (f supported near x_O). Along K_{v_1}, $\sigma = -2(v_1, x)$ and

$$\widehat{\Delta}^{(1)}(K_{v_1}) = -2(4-D)\int \mathrm{dvol}_D(v_1,x)\mathscr{L}_0, \qquad \widehat{\Delta}_O(K_{v_1}) = -2(4-D)\int \mathrm{dvol}_D(v_1,x) f\Phi,$$

plus the D-dependent gauge-fixing vertex $2(D-2)\int(\bar{c}, \iota_{v_1}A)$ in b_D. Writing $x = x_O + y$,

$$K_1(O) = -(v_1, x_O)\, D_1(O) + K_1^0(O),$$

where $K_1^0(O)$ is (3.5) with the insertions weighted by $-2(4-D)(v_1, y)$, i.e. by the first y-moment of the inserted vertex, computed as $v_1 \cdot \partial_q$ at $q = 0$ of the graphs with $\mathscr{L}_0$ (resp. $f\Phi$) injecting momentum q (with the factor $\pm i$ of the Fourier convention of [HK, ch. 1]), and with the term $-2\,\mathrm{LIM}\, b_1$ of the gauge-fixing vertex; its two-body kernel is the special conformal anomaly. Under the hypothesis that the one-loop graphs with the q-derivative of a zero-momentum insertion are infrared finite at generic external momenta (checked by gate G9b), K_1^0 is local, two-body plus one-body, and independent of any infrared prescription. The component $(K_{v_1}, P_{v_1'})$ of $L \circ L = 0$ gives $[\delta(K_{v_1}), \rho_0(P_{v_1'})] = (v_1, v_1')\delta(\Delta)$ on cohomology, since $\delta(P) = \delta(M) = 0$ (Corollary 3.21).

Proof. Theorem 2.17 (ii), Definition 3.19 and Proposition 2.13 (iii). The decomposition: $(v_1, x) = (v_1, x_O) + (v_1, y)$; the first part reproduces the dilatation computation of Theorem 3.22 with the factor $-(v_1, x_O)$ ($\sigma_\Delta = 2$, $\sigma_K = -2(v_1, x)$). A position factor y^μ at a vertex is $\pm i\,\partial/\partial q_\mu$ of the momentum injected there. Residues are polynomial in the momenta, so the q-derivative at $q = 0$ of the residue is defined; infrared poles at $D = 4$ would come from doubled lines with vanishing momentum, absent at generic momenta under the stated hypothesis. The last statement: [HK, cor:c12-rel (R5)]. □

Proposition 3.25 (Infrared safety of residues)**.** The residue $[\cdot]_{(4-D)^{-1}}$ of a one-loop graph is the coefficient of the ultraviolet region of its Schwinger integral and is a polynomial in the masses and momenta; adding a common mass m^2 to all lines changes it by a polynomial in m^2 whose m^2-independent part is the massless residue.

Proof. Theorem 4.2: the residue is $\int \mathcal{F}_0\, d\alpha$ with $\mathcal{F}_0$ polynomial in the momentum form, which becomes momentum form $+m^2$. □

3.7 Supercharges

Proposition 3.26 (Q_1 and S_1 in DR)**.** Q_1 is (3.5) along ξ with $\widehat{\Delta}^{(1)}(\xi) = \mathfrak{F}(\xi)$ (Scheme D) or $\mathfrak{F}^{\mathrm{B}}(\xi) + \mathcal{E}(\xi)$ (Scheme B) on the cycles; $\widehat{\Delta}_O(\xi) = 0$ for lifts that transform letters by $S_D^{(1)}(\xi)$. Along η, since $S_D^{(1)}(\eta) = \sum_\mu \int x^\mu S_D^{(1)}(\rho(e_\mu)\eta)_x - \int(\rho(A|_E)\eta, \lambda^*)$ holds exactly in D dimensions,

$$S_1(O) \text{ along } \eta = Q_1(O) \text{ along } \rho(x_O)\eta + S_1^0(O),$$

with S_1^0 from the y-moment of the Fierz vertex, the term $(4-D)\mathcal{N}(\eta)$ and the cycles of $-(\rho(A|_E)\eta, \lambda^*)$. The gauge-fixing λ–$\bar{c}$ vertex of Proposition 2.13 (ii) enters through the arity-one counterterm $\partial_\delta I_D^{CT,(1)}(O)(\xi)$ in $\mathcal{C}_\xi$.

Proof. Theorem 3.20, Theorem 2.17 (iii)–(iv), and the position decomposition of $S^{(1)}(\eta)$ [HK, def:c12-gens] transcribed by T2–T3. □

3.8 Scheme laws and invariant content

Proposition 3.27 (Laws in DR)**.** Under the following changes L_1 changes as stated:

(a) field redefinition $F \mapsto F + \hbar s\phi$: $L_1 \mapsto L_1 - \mathrm{ad}_{s\phi}$ (conjugation by $1 + \hbar\,\mathrm{ad}_\phi$);
(b) finite parts $\psi \mapsto \psi + \psi'$, including MS $\to$ MS-bar and evanescent changes of the lifts O_D: $L_1 \mapsto L_1 + [s, \psi']$;
(c) closed directions $F \mapsto F + \hbar aG$, $sG = 0$: $L_1 \mapsto L_1 - a\,\mathrm{ad}_G$, with the cases $G = S - S^{(2)}$, $G = G_B$ of [HK, prop:c8-laws];
(d) the gauge parameter: (a) by Theorem 3.13;
(e) evanescent finite counterterms ($\mathrm{LIM}\, F^{\mathrm{ev}} = 0$) and evanescent lifts of $S_D^{(k)}$, Ψ: no change at one loop up to (a); a change of the two-loop data;
(f) the choice of scheme (D or B, J in Scheme B): two one-loop quantisations of the same four-dimensional theory, related by (a)–(c).

Proof. (a)–(c) as [HK, prop:c8-laws], the proofs using only (3.4), the derivation rules of the bracket and Lemma 3.16. (d) Theorem 3.13. (e) Theorem 3.11 (iv) and Lemma 3.10: $\mathrm{LIM}[E\cdot\Gamma]^{(1)}$ is a finite local cochain, absorbed in F_1 or ψ. (f) Both Γ_{ren} solve $S(\Gamma_{\mathrm{ren}}) = O(\hbar^2)$ with the same tree level; their difference at order $\hbar$ is an s-closed local cochain modulo field redefinitions and finite parts. □

Definition 3.28 (What the DR computation must reproduce)**.** With the hypotheses (H0) (local representatives on the families), (Hs) ($H^0(s) = \mathbb{C}[S - S^{(2)}] \oplus \mathbb{C}[G_B]$ at weight 4) and (Hc) of [FR], the invariant one-loop data are:

(I1) the dilatation matrix $\delta(\Delta)$ at a point, i.e. D_1 at the origin on every family: the whole mixing matrix, its eigenvectors and eigenvalues [FR, Thm 4.1];
(I2) the twisted one-loop differential $d_1 = [Q_1^\xi]$ on $E_1 = H(Q_0^\xi)$, $Q_1^\xi = L_1^{(0)} + L_1^{(1)}(\xi) - \frac{1}{2}L_1^{(2)}(\xi,\xi) + \cdots$, defined modulo $[Q_0^\xi, h] + \mathbb{C}\,Q_0^\xi$, and its reduced pair entries [FR, Thm 5.1, Thm 8.1];
(I3) for generic Q, S, K: the class $[\delta] \in H^1(\mathfrak{psl}(4|4); \mathrm{End}\,\mathbb{V})/\mathbb{C}[\rho_0 \circ B]$, i.e. δ modulo $[\rho_0, e]$ with one weight-preserving e shared by all generators; on simple summands $\rho_0 \circ B$ is itself a coboundary [FR, Thm 8.1, Prop 4.3];
(I4) on non-split extensions $0 \to N \to E \to C \to 0$ (joining of multiplets, the Konishi case): the lifting class $[\kappa]$, the identity $\kappa(\eta)\beta(\xi)c_0 = -\gamma(\xi,\eta)c_0$ and, in a general direction, $(a_\xi + a'_\eta)\mu c_0 + \kappa(\eta)\beta(\xi)c_0 = -\gamma(\xi,\eta)c_0$ with $a_\xi + a'_\eta$ invariant [FR, Thm 4.9]; the one-loop norms $\|\rho(\xi)c_0\|^2_{\mathrm{rad}} = (\xi,\bar\xi)\gamma\|c_0\|^2_{\mathrm{rad}}$ [FR, Cor 4.10];
(I5) protection: on a simple summand with $\gamma = 0$ every one-loop component is one coboundary $[\rho_0, e]$ [FR, Thm 4.7], in particular on the $\mathbf{20'}$ bilinears;
(I6) P, M, R: pure gauge on simple summands [FR, Thm 8.1].

Theorem 3.29 (DR reproduces the invariants; comparison of the tracks)**.** Assume (H0), (Hs), (Hc). Scheme D and Scheme B, each with its quantum master equation restored by the finite counterterms of Theorem 3.12, are regularisations in the sense of [FR, Thm 2.3]: relative to any other such regularisation with the same valid gauge fixing, $\mathrm{Ob} \mapsto \mathrm{Ob} + s\Gamma$, $F \mapsto F + \Gamma$, $\psi \mapsto \psi + \psi_{RR'}$, and the set of one-loop structures L_1 is the same. Hence the data (I1)–(I6) computed in Scheme D, in Scheme B and in the heat-kernel track on a valid gauge fixing coincide. In DR, (I6) holds at the level of representatives: $P_1 = M_1 = 0$, and $R_1 = 0$ in Scheme D (Corollary 3.21).

Proof. Theorem 2.3 of [FR] applies: its proof uses Lemma [FR, Lem. locality of scheme differences], which holds for DR by continuity in D at $D = 4$ of the convergent derivatives $\partial_p^{\deg+1} G$, and the injectivity of the tree flow, Theorem 3.18 (i). The gauge fixing of Definition 2.9 is valid in the sense of [FR, Thm 2.4]: $(Q^{GF})^2 = 0$, $[Q, Q^{GF}] = H$, covariance under translations, $\mathfrak{so}(4S)$, $\mathfrak{so}(W^\perp)$ and scaling ($\bar c$ of weight 2), Lemma 2.10. The invariance of (I1)–(I6) under these changes is [FR, Thm 4.1, 4.7, 4.9, 5.1, 8.1, Prop 4.3, Cor 4.10]. The last statement is Corollary 3.21. □

Remark 3.30 (Where the twisted Konishi entry sits in DR)**.** In the heat-kernel track the Konishi entry of d_1 is carried by $\hbar L_1^{(0)}$ [FR, §5]. In DR, $\hbar L_1^{(0)} = 0$ (Corollary 3.21), so the same class is carried by $L_1^{(1)}(\xi) - \frac{1}{2}L_1^{(2)}(\xi,\xi)$: the arity-one Q_1 and the arity-two kernel along (ξ,ξ). In particular $Q_1 \neq 0$ on the Konishi multiplet in every DR scheme with restored master equation, by (I2) and (I4) with $\gamma_K \neq 0$.

4 One loop: two letters, closed formulas and evanescent operators

At one loop a graph meets a local operator in at most two letters: its cycle uses two, a tree uses one [HK, thm:c10-decomp] (the R8 structure of [HK, DESIGN_STEP2]). In DR the pair kernels have no power divergences and no self-contraction term, their residues are simplex integrals of a polynomial, and the finite parts of the evanescent insertions that produce every one-loop component are derivatives in D of these residues (Lemma 3.10).

4.1 The decomposition

Theorem 4.1 (Two-letter decomposition in DR)**.** For a monomial $M = \int \omega\, X_1 \cdots X_n$,

$$\partial_\delta I_D^{CT}(M) = \sum_{1\le a<b\le n} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1 \cdots X_{a-1}\, \partial_\delta I_D^{CT}(X_a X_b)\, X_{a+1} \cdots \widehat{X_b} \cdots X_n,$$

with the pair kernel $\partial_\delta I_D^{CT}(X_aX_b)$ the pole part of $\hbar\,\partial_x^\alpha \partial_y^\beta$ of the value of $\partial_{\text{dressed propagator}}$ on $\varphi_a(x)\varphi_b(y)$ at $y = x$, the dressed propagator built from P_D and the one-loop operator of I_Ψ [HK, def:c10-pair]. The term without vertices vanishes; every pair kernel is $(4-D)^{-1}$ times a local density, of output weight equal to the pair weight $w_{X_a} + w_{X_b}$. The same holds with $I_{\Psi,\mathfrak{psl}}$, giving the arity-one and arity-two counterterms, of output weight $w_{X_a} + w_{X_b} - \frac{1}{2}\sum_i \lambda_{x_i}$.

Proof. The combinatorics of [HK, thm:c10-decomp (i)–(iii)] is regulator-independent: the δ-linear one-loop graphs through M are trees of I on the legs of $\hbar\,\partial_{\text{dressed propagator}} M$, and ∂_ϕ is a second-order operator. The term without vertices is P_D at coinciding points with derivatives, $\int d^Dp\, p^\alpha p^\beta |p|^{-2} = 0$. One-loop integrals in DR have only simple poles and no log; dimensional analysis with a log-divergent residue fixes the output weight. □

4.2 The cycle formula

Theorem 4.2 (One-loop cycles in D dimensions)**.** For a chain with k vertices and $k+1$ edges carrying $l + q_j$ ($q_0 = 0$), numerator $N(l)$ (polynomial in l and the output momenta, with the fibre and colour factors), centre $c = -\sum_j \alpha_j q_j$ and momentum form $\mathfrak{m} = \sum_j \alpha_j |q_j|^2 - |\sum_j \alpha_j q_j|^2$ on the simplex with the Lebesgue measure $d\alpha$ of $(\alpha_1,\dots,\alpha_k)$:

$$\int \frac{d^Dl}{(2\pi)^D}\, N(l) \prod_{j=0}^{k} \frac{1}{|l+q_j|^2} = (4\pi)^{-D/2} \int d\alpha \sum_{j\ge 0} \frac{((\sum_\mu \partial_{l_\mu}^2)^j N)(c)}{j!\,4^j}\, \Gamma\Big(k+1-\tfrac{D}{2}-j\Big)\, \mathfrak{m}^{\frac{D}{2}+j-k-1}. \tag{4.1}$$

With $n := j-k+1$ and $\mathcal{F}_0 := \sum_{j\ge k-1} \frac{((\sum \partial_l^2)^j N)(c)}{j!\,4^j} \frac{(-\mathfrak{m})^n}{n!}$,

$$16\pi^2\, [\text{cycle}]_{(4-D)^{-1}} = 2\int \mathcal{F}_0\, d\alpha, \tag{4.2}$$

$$\begin{aligned} 16\pi^2\, [\text{cycle}]_{(4-D)^0} = \int d\alpha \Big\{ &\sum_{j\ge k-1} \frac{((\sum \partial_l^2)^j N)(c)}{j!\,4^j} \frac{(-\mathfrak{m})^n}{n!} (\psi(n+1) + \log 4\pi - \log \mathfrak{m}) \\ &+ \sum_{j<k-1} \frac{((\sum \partial_l^2)^j N)(c)}{j!\,4^j}\, (k-2-j)!\, \mathfrak{m}^{j+1-k} \Big\}, \end{aligned} \tag{4.3}$$

ψ the digamma function. Every residue is a polynomial in the output momenta with rational coefficients, by $\int \prod_e \alpha_e^{n_e} d\alpha = \prod_e n_e!/(\sum_e n_e + k)!$.

Proof. $\prod_j |l+q_j|^{-2} = \int_{[0,\infty)^{k+1}} \prod dt_j\, e^{-\sum t_j|l+q_j|^2}$, $t_j = \tau\alpha_j$, $\prod dt_j = \tau^k d\tau\, d\alpha$, $\sum_j t_j|l+q_j|^2 = \tau(|l-c|^2+\mathfrak{m})$. The Gaussian integral with a polynomial numerator is $(4\pi\tau)^{-D/2}\sum_j((\nabla_l^2)^j N)(c)/(j!(4\tau)^j)$ (Axiom 2.1; $\nabla_l^2 = \sum_\mu \partial_{l_\mu}^2$ with $\partial_{l_\mu} l_\mu$ summed to D over QDS). Then $\int_0^\infty d\tau\, \tau^{k-D/2-j} e^{-\tau\mathfrak{m}} = \Gamma(k+1-\frac{D}{2}-j)\mathfrak{m}^{\frac{D}{2}+j-k-1}$. For $n \geq 0$, $\Gamma(-n+\frac{4-D}{2}) = \frac{(-1)^n}{n!}(\frac{2}{4-D}+\psi(n+1)) + O(4-D)$, $(4\pi)^{-D/2} = \frac{1}{16\pi^2}(1+\frac{4-D}{2}\log 4\pi + \cdots)$, $\mathfrak{m}^{n-\frac{4-D}{2}} = \mathfrak{m}^n(1-\frac{4-D}{2}\log\mathfrak{m}+\cdots)$; for $n<0$ the terms are finite at $D=4$. □

Remark 4.3. In Scheme D the tensor algebra of N uses QDS-contractions for loop momenta ($l_\mu l_\mu \to$ trace D) and $Q4S \oplus W^\perp$-contractions for field indices (trace 10); in Scheme B the loop momentum splits as $l = l|_{4S} + l|_{Q(D-4)S}$ and ∇_l^2 produces the traces 4 and $D-4$ separately. These D-dependences of N are the polynomial D-dependence of $\mathcal{R}_X(D)$ in Lemma 3.10; the numerator must be kept in normal form before LIM.

Proposition 4.4 (Generating functions)**.** For the numerator $e^{i(y,l)}$, $y = y_b - y_a$,

$$\int \frac{d^D l}{(2\pi)^D} \frac{e^{i(y,l)}}{\prod_j |l+q_j|^2} = (4\pi)^{-D/2} \int d\alpha\, e^{i(y,c)}\, 2\Big(\frac{|y|^2}{4\mathfrak{m}}\Big)^{\frac{k+1-D/2}{2}} K_{k+1-\frac{D}{2}}(|y|\sqrt{\mathfrak{m}}),$$

and the integrand of (4.2) is

$$\mathcal{F}_0 = (-1)^{k-1} e^{i(y,c)} \sum_{i\geq 0} \frac{(|y|^2/4)^{i+k-1}\mathfrak{m}^i}{i!\,(i+k-1)!} = (-1)^{k-1} e^{i(y,c)} \Big(\frac{|y|}{2\sqrt{\mathfrak{m}}}\Big)^{k-1} I_{k-1}(|y|\sqrt{\mathfrak{m}}),$$

zero for $k=0$. With the letters at $x+y_a$, $x+y_b$ and a chain numerator $N_{\mathrm{ch}}(l)$, the generating function of the residues of the pair kernels is

$$\sum_{\alpha,\beta} \frac{y_a^\alpha y_b^\beta}{\alpha!\beta!} \left[\partial_\delta I_D^{CT}(\partial^\alpha\varphi_a \partial^\beta\varphi_b)\right]_{(4-D)^{-1}} = \frac{2\hbar}{16\pi^2}\, e^{i(y_b,q_k)} \int N_{\mathrm{ch}}(-i\partial_y)\, \mathcal{F}_0\, d\alpha,$$

summed over the chains; in position space iq_j is the derivative acting on the product of the outputs at x.

Proof. $\int d^D l\, e^{i(y,l)} e^{-\tau|l-c|^2} = (\pi/\tau)^{D/2} e^{i(y,c)} e^{-|y|^2/4\tau}$ and $\int_0^\infty d\tau\, \tau^{\nu-1} e^{-\tau\mathfrak{m}-|y|^2/4\tau} = 2(|y|^2/4\mathfrak{m})^{\nu/2} K_\nu(|y|\sqrt{\mathfrak{m}})$ with $\nu = k+1-\frac{D}{2}$. For the residue use (4.2) with $(\nabla_l^2)^j e^{i(y,l)} = (-|y|^2)^j e^{i(y,l)}$, $j = i+k-1$; the letters carry $e^{-i(y_a,l)+i(y_b,l+q_k)}$. □

Corollary 4.5 (The bubble in closed form)**.** For $k=1$ and a constant chain numerator, the pair $\varphi_a(x+y_a)\varphi_b(x+y_b)$ has the counterterm

$$\frac{2}{4-D}\,\frac{\hbar}{16\pi^2} \sum_{n_1,n_2,j\geq 0} \frac{(n_1+j)!\,(n_2+j)!}{n_1!\,n_2!\,(j!)^2\,(n_1+n_2+2j+1)!}\,(y_a,\partial)^{n_1}(y_b,\partial)^{n_2}\,|y_b-y_a|^{2j}\big(-\tfrac{1}{4}\partial^2\big)^j$$

acting on the product of the two outputs, times the colour and fibre factor of the chain. At $n_1 = n_2 = j = 0$ it is $\frac{2}{4-D}\frac{\hbar}{16\pi^2}$; for one derivative on each letter the coefficient of $y_a^\mu y_b^\nu$ is $\frac{2}{4-D}\frac{\hbar}{16\pi^2}(\frac{1}{6}\partial_\mu\partial_\nu + \frac{1}{12}\delta_{\mu\nu}\partial^2)$ on the outputs.

Proof. $k=1$: $c = -\alpha q_1$, $\mathfrak{m} = \alpha(1-\alpha)|q_1|^2$, $\mathcal{F}_0 = e^{-i\alpha(y,q_1)}\sum_j (|y|^2|q_1|^2\alpha(1-\alpha)/4)^j/(j!)^2$; with the prefactor $e^{i(y_b,q_1)}$ the exponent is $i(1-\alpha)(y_b,q_1) + i\alpha(y_a,q_1)$, and $\int_0^1 \alpha^{n_1+j}(1-\alpha)^{n_2+j}d\alpha = \frac{(n_1+j)!(n_2+j)!}{(n_1+n_2+2j+1)!}$. With one derivative on each letter: $(n_1,n_2,j) = (1,1,0)$ gives $\frac{1}{3!}(y_a,\partial)(y_b,\partial)$, and $(0,0,1)$ gives $\frac{1}{3!}|y_b-y_a|^2(-\frac{1}{4}\partial^2)$, whose $y_a^\mu y_b^\nu$-coefficient is $\frac{1}{3!}(-2\delta_{\mu\nu})(-\frac{1}{4}\partial^2) = \frac{1}{12}\delta_{\mu\nu}\partial^2$. □

Definition 4.6 (Defect cycles in DR)**.** For a generator x and a pair (X_a, X_b), the DR defect cycles are the chains from X_a to X_b with vertices of $I_{\Psi,\mathfrak{psl}}$, edges P_D and exactly one vertex taken from $\widehat{\Delta}^{(1)}(x)$ (Theorems 2.17, 2.18); their residue polynomial in normal form is $\mathcal{R}^{\mathrm{cyc}}_x(X_a, X_b)(D)$ of Theorem 3.20. Placements: the vertex of $\widehat{\Delta}^{(1)}(x)$ sits on the cycle with two of its legs; for $x = \Delta, K_{v_1}$ its legs are those of $\mathscr{L}_0$ (a kinetic or interaction vertex of $S_{0,D}$, weighted by $(4-D)\sigma_v$); for $x = \xi$ in Scheme D it is the cubic Fierz vertex $\mathfrak{F}(\xi)$ with two λ-legs on the cycle and one output λ; in Scheme B also the quadratic and cubic vertices of $\mathcal{E}(\xi)$.

Proposition 4.7 (Active pairs and output bound)**.**

(i) The active slots are those with a nonzero component of P_D: $A|_{QDS}$, $A|_E$, λ, c, $\bar{c}$, b and, through Q^{GF}, none of the antifields; antifield letters are spectators of every pair kernel.

(ii) For a pair of total weight w and the generator x, the residue and Υ^D_x have output weight $w - \frac{1}{2}\lambda_x$; a defect cycle with more output fields than this weight allows is finite and has $\mathrm{LIM} = 0$.

(iii) Along Δ and K_{v_1} the defect vertex is $(4-D)\sigma_v$ times a vertex of $\mathscr{L}_0$, so Υ^D_Δ, $\Upsilon^D_{K_{v_1}}$ are, up to the factors $-2\partial_D(4-D) = 2$ and the position weights, residues of ordinary cycles with one additional action vertex: no evanescent algebra is needed along Δ and K beyond that of the counterterms.

(iv) Along ξ in Scheme D every defect cycle contains the Fierz vertex once; Υ^D_ξ is $-\partial_D$ of the D-dependence created by the QDS-contractions of the loop momentum with ρ's adjacent to the Fierz vertex.

Proof. (i) Lemma 2.10: Q^{GF} reads antifields, so P_D has components only between field slots. (ii) Theorem 4.1 and the weight of $\widehat{\Delta}^{(1)}(x)$, $4 - \frac{1}{2}\lambda_x$ (relative to D). (iii) Theorem 2.17 (ii) and Lemma 3.10 with $\mathcal{R}_X(D) = (4-D)\mathcal{R}_{\mathscr{L}_0}(D)$. (iv) The contractions of the field indices of the Fierz vertex are over $Q4S \oplus W^\perp$ (trace 10, D-independent); the only D comes from ∇_l^2. □

4.3 Evanescent operators

Definition 4.8 (Evanescent operators and the evanescent scheme)**.** An evanescent operator is a local functional E in normal form with $\mathrm{LIM}\, E = 0$: in Scheme B a monomial with an uncontracted component along $Q(D-4)S$, a factor ρ_J, or a contracted evanescent trace; in Scheme D a Fierz-null combination, for instance $\rho \circ q$ of more than four $Q4S$-directions, the difference between an antisymmetrised product of three $\rho(Q4S)$ and its four-dimensional dual, or the cubic structure of $\mathfrak{F}$. An evanescent scheme is a choice, for each normal-form residue $\mathcal{R}$, of a decomposition $\mathcal{R} = \mathcal{R}^{\mathrm{phys}}_D + E$ with $\mathcal{R}^{\mathrm{phys}}_D$ a fixed D-dimensional lift of $\mathrm{LIM}\,\mathcal{R}$; two schemes differ by $(4-D)\times$(local) in $\mathcal{R}^{\mathrm{phys}}_D$, e.g. by the coefficient b in $\rho(q(e^\mu \wedge e^\nu \wedge e^\sigma)) \otimes \rho(q(e_\mu \wedge e_\nu \wedge e_\sigma)) = (a + b(4-D))\, \rho(e^\mu) \otimes \rho(e_\mu) \cdot (\mathrm{chirality}) + E$.

Proposition 4.9 (Evanescent operators at one and two loops)**.**

(i) The one-loop counterterms of local operators and of the action contain evanescent pole terms $(4-D)^{-1}E$, required for the finiteness of the D-dimensional theory; in Scheme D they arise in pair kernels with two spinor letters (open spinor chains whose four-dimensional reduction uses Fierz identities), in Scheme B in addition from the hat components of loop momenta.

(ii) $\mathrm{LIM}[E \cdot \Gamma_D]^{(1)}$ is a finite local physical cochain (Lemma 3.10); adding the finite counterterm $-\mathrm{LIM}[E \cdot \Gamma_D]^{(1)}$ to E makes the renormalised evanescent operators vanish in four dimensions at one loop [DG91], [HN95].

(iii) The one-loop components L_1 on physical operators are independent of the evanescent scheme: a change $(4-D) \cdot e$ of $\mathcal{R}^{\mathrm{phys}}_D$ changes the counterterm by the finite e, i.e. by law (b).

(iv) At two loops the evanescent scheme and the finite counterterms of (ii) enter $\mathrm{Ob}^{(2)}$, F_2 and $\gamma^{(2)}$; when $\beta = 0$ the change of $\gamma^{(2)}$ is a commutator with $\gamma^{(1)}$, so the two-loop spectrum is unchanged.

Proof. (i) Theorem 4.2 in normal form: products of ρ's along the open chain are reduced by the Clifford relation only. (ii), (iii) Lemma 3.10: $(4-D)^{-1} \cdot (4-D)e = e$ is finite and local. (iv) [DG91], [HN95]: the two-loop anomalous-dimension matrix changes by $[\gamma^{(1)}, r]$ plus terms

proportional to β. □

Remark 4.10 (Closed fermion loops and the chirality-odd traces)**.** In Scheme D the chirality-odd part of every trace is set to 0 (Definition 2.3); in strict ten dimensions it is nonzero only for a product of at least ten ρ's. A closed fermion loop with n_v vertices of type $\lambda\lambda A$ or Yukawa carries $2n_v$ factors ρ, and its superficial degree is $4 - n_v$; the loops with a pole have $n_v \leq 4$, at most eight ρ's, and the loops with $n_v \geq 5$ are finite. Pair kernels have no closed fermion loop through the operator. Hence the chirality-odd traces affect no pole and no LIM of an evanescent insertion at one loop, and they affect the finite parts of one-loop correlators with at least five vertices on a fermion loop (parity-odd terms of five-point functions).

5 Verification of the algebra

The relations among the one-loop components are the arity-two components of $L \circ L = 0$; they hold on the cohomology of $\Lambda^{(0)}$ as soon as the quantum master equation holds through arity two. Without restoring counterterms they fail by the unrestored obstruction, which makes each relation a computable test of the scheme.

Theorem 5.1 (Relations and their defect)**.** Let $S(\Gamma_{\mathrm{ren}}) = \hbar\,\mathcal{B} + O(\hbar^2)$ with $\mathcal{B} := \mathrm{Ob} - sF_1$. Then $S_\Gamma^2 = \hbar\,\{\mathcal{B}, -\} + O(\hbar^2)$ on renormalised insertions, and on local operators

$$L \circ L = \hbar\,\mathrm{ad}_{\mathcal{B}} + O(\hbar^2).$$

With $\delta :=$ the order-$\hbar$ part of $-\Lambda^{(1)}$ on cohomology and ρ_0 the classical action [HK, def:c12-class], the arity-two component at order $\hbar$ on cocycles O reads

$$\begin{aligned}[\delta(x), \rho_0(y)] + [\rho_0(x), \delta(y)] - \delta([x,y]) = {} & \pm \{\mathcal{B}^{(2)}(x,y), -\} \pm [\{\mathcal{B}^{(1)}(x), -\}, \rho_0(y)] \\ & \pm [\rho_0(x), \{\mathcal{B}^{(1)}(y), -\}] + \Lambda^{(0)}\text{-exact},\end{aligned} \tag{5.1}$$

with the signs of the cochain bracket [HK, def:c2-dce]. If $sF_1 = \mathrm{Ob}$ through arity two along (x, y), the right side is $\Lambda^{(0)}$-exact and the relations (R2)–(R8) of [HK, cor:c12-rel] hold on cohomology.

Proof. For any functional Γ, $S_\Gamma^2 = (S(\Gamma), -)$ by the Jacobi identity and the derivation property of d_{CE}. Applied to $[M]_R$ and (3.4), $[L(L(M))]_R = S_\Gamma^2[M]_R = \hbar(\mathcal{B}, [M]_R) + O(\hbar^2)$, and by Theorem 3.18 (i) $L \circ L = \hbar\,\mathrm{ad}_{\mathcal{B}}$ at this order. The arity-two component is [HK, cor:c8-homotopy (ii), thm:c12-opalg] with this right side. □

Definition 5.2 (The relations to verify)**.** On cohomology, with $\delta(P_{v_1}) = \delta(M_\omega) = 0$ in DR (Corollary 3.21):

(R2) $\{Q, Q\} = P$: $[\delta(\xi), \rho_0(\xi')] + [\rho_0(\xi), \delta(\xi')] = \delta(P_{\Gamma(\xi,\xi')|_V}) = 0$.

(R3) $\{Q, S\} \supset D$: $[\delta(\xi), \rho_0(\eta)] + [\rho_0(\xi), \delta(\eta)] = \frac{1}{2}(\xi, \eta)\,\delta(\Delta) + \delta(R_{\xi\eta})$, $R_{\xi\eta}$ the $\mathfrak{so}(W^\perp)$-part of $[\xi, \eta]$ [HK, prop:c2-special].

(R4) rank-one $\xi \in S_+ \otimes W$: $\{\delta(\xi), \rho_0(\xi)\} = 0$, so the twisted class $[Q_1^\xi]$ is defined [HK, thm:c8-twist].

(R5) $[K, P] \supset D$: $[\delta(K_{v_1}), \rho_0(P_{v_1'})] = (v_1, v_1')\,\delta(\Delta)$.

(R6) $\{S, S\} = K$: $[\delta(\eta_1), \rho_0(\eta_2)] + [\rho_0(\eta_1), \delta(\eta_2)] = \delta([\eta_1, \eta_2])$.

(R7) $[D, x]$: $[\delta(\Delta), \rho_0(x)] = 0$ on a family at one point graded by the classical weight, for every generator x; for $x = \xi$ this is $[D, Q]$: D_1 is constant on classical supermultiplets.

(R9) $[K, Q] = S$: $[\delta(K_{v_1}), \rho_0(\xi)] + [\rho_0(K_{v_1}), \delta(\xi)] = \delta(\rho(v_1)\xi)$; $[P, S] = Q$: $[\rho_0(P_{v_1}), \delta(\eta)] = \delta(\rho(v_1)\eta)$, which is the position decomposition of Proposition 3.26.

(R10) the Konishi relation: (R3) on the trace bilinear $\sum_I (A_I, A_I)$ and its first descendants, whose right side is $\frac{1}{2}(\xi, \eta)D_1 = -24(\xi, \eta)$ units.

(R11) joining and lifting: on a non-split extension $0 \to N \to E \to C \to 0$ with classical joining cocycle β and one-loop lifting κ, $\kappa(\eta)\beta(\xi)c_0 = -\gamma(\xi, \eta)c_0$ [FR, Thm 4.9 (iv)]; in the $\mathfrak{su}(2|3)$ sector this is the closure relation $c_9 c_{10} = \frac{1}{2}c_1$ of [Bei03b, (3.36)] between the coefficients of the length-changing supercharges, read from the tree-level BV data, and the one-loop dilatation normalisation c_1 [FR, §7].

Theorem 5.3 (Exactness table)**.** In the minimal scheme ($\psi = 0$) the relations hold on cohomology as follows; "MS" means with $F_1 = 0$ in the components listed, "F" with the restoring counterterms of Theorem 3.12, "test" means that the vanishing of the defect (5.1) with $F_1 = 0$ is a prediction to be computed.

relation	obstruction components entering	Scheme D	Scheme B
gauge BRST, (R1)	$\mathrm{Ob}^{(0)}$	MS	MS
P, M covariance	$\mathrm{Ob}^{(1)}(P), \mathrm{Ob}^{(1)}(M)$	MS	MS
R covariance	$\mathrm{Ob}^{(1)}(R_r)$	MS	MS on $\mathfrak{so}(4)_J$; F otherwise
(R7) along P, M, R	as above	MS	as above
(R2) $\{Q, Q\} = P$	$\mathrm{Ob}^{(1)}(\xi)$, $\mathrm{Ob}^{(2)}(\xi, \xi')$	test (Fierz only)	F (field counting)
(R4) twisted	$\mathrm{Ob}^{(1)}(\xi)$, $\mathrm{Ob}^{(2)}(\xi, \xi)$	test	F
(R7) $[D, Q]$	$\mathrm{Ob}^{(1)}(\Delta)$, $\mathrm{Ob}^{(1)}(\xi)$	F along Δ (field counting); test along ξ	F
(R3) $\{Q, S\} \supset D$, (R10)	$\mathrm{Ob}^{(1)}(\xi, \eta, \Delta)$, $\mathrm{Ob}^{(2)}(\xi, \eta)$	F (the $(4-D)\mathcal{N}(\eta)$ and Δ parts)	F
(R5) $[K, P] \supset D$	$\mathrm{Ob}^{(1)}(K), \mathrm{Ob}^{(1)}(\Delta)$	F along Δ, K	F
(R6) $\{S, S\} = K$	$\mathrm{Ob}^{(1)}(\eta)$, $\mathrm{Ob}^{(2)}(\eta, \eta')$	F	F
(R9)	$\mathrm{Ob}^{(1)}(K, \xi, \eta)$, $\mathrm{Ob}^{(2)}(K, \xi)$	F	F
(R11) joining/lifting	as (R3)	as (R3)	as (R3)

The entries "F" along Δ, K and the $(4-D)\mathcal{N}(\eta)$-part along η are not symmetry violations: they carry the anomalous dimensions of the elementary fields and the conformal anomaly of the regulated action, and the corresponding F_1 are field-counting functionals and their position moments. The entries "test" in Scheme D are the prerequisite and part (a) of Problem 7.2.

Proof. Theorem 5.1 with Theorem 3.12 and Corollary 3.21. Along ξ in Scheme D, b_D has no explicit D-dependence (neither $S_D^{(1)}(\xi)$ nor $\xi.\Psi$ contains D), so by Theorem 3.11 (i) $\mathrm{Ob}^{(1)}(\xi) = -\mathrm{LIM}\,\partial_D \mathcal{R}_{\mathfrak{F}(\xi)}$, a pure Fierz-insertion term; the same holds for $\mathrm{Ob}^{(2)}(\xi, \xi')$ with the Fierz-null $\widehat{\Delta}^{(2)}(\xi, \xi')$ of Theorem 2.17 (v). In Scheme B, $\mathcal{E}(\xi)$ contributes already to the two-leg part, as for γ_5 of 't Hooft–Veltman type in [S05, (4.5)]. □

Definition 5.4 (Verification protocol)**.** For each relation of Definition 5.2 and each family $\mathfrak{F}$ (scalar bilinears; the trace bilinear and its descendants of weight $\frac{5}{2}$ and 3; the $\mathfrak{su}(2)$ sector of two-letter words; twist two at lowest spin):

1. compute $\delta(x)$ for the generators involved from (3.5) (two-body kernels Υ_x^D, one-body terms, $\{F_1^{(1)}(x), -\}$);
2. compute the arity-two kernels $\Upsilon_{x,y}^D$ (cycles through the pair with two arity-one vertices or one arity-two vertex, evanescent vertex once) and $\chi_{x,y}$ of [HK, thm:c12-opalg (ii)];
3. evaluate the left side of (5.1) on a basis of $\mathfrak{F}$ as an exact matrix identity, subtract the $\Lambda^{(0)}$-exact part $\Lambda^{(0)}\chi_{x,y}$, and compare the remainder with the right side computed from $\mathcal{B}$;
4. repeat with $F_1 = 0$ (MS) and with the restoring F_1; record which relations hold in which scheme, and the scheme-independent combinations (classes, spectra).

Route A of Section 6 performs 1–4 with residues; Route B evaluates the same relations on renormalised Green functions with operator insertions, without the quantum action principle.

Remark 5.5 (Two-component form of the relations)**.** With Q^i_α, $\bar{Q}_{\dot\alpha i}$, $S^\alpha{}_i$, $\bar{S}^{\dot\alpha i}$ labelled by the components $\xi_\alpha{}^i$, $\bar{\xi}_{\dot\alpha i}$, $\eta^\alpha{}_i$, $\bar{\eta}^{\dot\alpha i}$ [HK, def:c8-twocomp] and the brackets of [HK, prop:c2-special]:

- (R2): the one-loop parts of $\{Q^i_\alpha, \bar{Q}_{\dot\beta j}\}$ and $\{Q^i_\alpha, Q^j_\beta\}$ vanish on cohomology, their right sides being $\delta(P) = 0$; the translation is $\Gamma(\xi, \bar{\xi}')|^m_V = i\,\bar{\xi}'_{\dot\alpha i}\bar{\sigma}^{m\dot\alpha\alpha}\xi_\alpha{}^i$.
- (R3): the dilatation part of $[\xi, \eta]$ is $\frac{1}{2}(\xi, \eta)\Delta$ with $(\xi, \eta) = \xi_\alpha{}^i\eta^\alpha{}_i - \bar{\xi}_{\dot\alpha i}\bar{\eta}^{\dot\alpha i}$, so the one-loop part of $\{Q^i_\alpha, S^\beta{}_j\}$ contains $\frac{1}{2}\delta^i_j\delta_\alpha{}^\beta D_1$, plus the $\mathfrak{su}(4)$ generator, whose one-loop part vanishes in Scheme D.

- (R5): $[K_m, P_n]$ at one loop is $\delta_{mn} D_1$, the rotation part being $M_1 = 0$.
- (R6): the one-loop part of $\{S^\alpha{}_i, \bar{S}^{\dot\beta j}\}$ is δ of $-K_{v_1}$ with $v_1 = \Gamma(\eta, \bar\eta')|_V$.

6 Engine specification

The machine-readable specification is `engine_spec.json`; this section records its logic. The targets are the invariants (I1)–(I6) of Definition 3.28 and the questions Q1–Q9 of Section 7, each listed there with its loop order.

Definition 6.1 (Route A: residues and the quantum action principle)**.** Route A computes every one-loop component from graphs with exactly one evanescent vertex: the vertices of $S_{\Psi,\mathfrak{psl}}$ and the components of $\widehat{\Delta}$ are generated symbolically in D (Theorems 2.17, 2.18); the pair cycles are enumerated within the output bound (Proposition 4.7); residues are simplex integrals (Theorem 4.2) with generating functions in the letter positions (Proposition 4.4); finite parts are $-\partial_D$ of normal-form residues at $D = 4$ (Lemma 3.10); $sF_1 = \mathrm{Ob}$ is solved by exact linear algebra with all free directions; the components follow from (3.5).

Definition 6.2 (Route B: direct evaluation)**.** Route B computes the one-loop one-particle-irreducible functions and the operator insertions completely in momentum space (tensor reduction to B_0, C_0 with exact D-dependence), from independently written two-component Feynman rules, takes LIM, and evaluates $S(\Gamma_{\mathrm{ren}})$ and $S_\Gamma[O]_R$ as sums and products of one-particle-irreducible functions; it uses neither $\widehat{\Delta}$ nor Lemma 3.10.

Remark 6.3 (Gates)**.**

gate	content
G0a–G0d	Clifford realisations (Scheme B explicit, Scheme D formal), hat algebra, equality of the two hand-written classical inputs
G1a–G1c	$Q^2 = 0$, $(Q^{GF})^2 = 0$, $[Q, Q^{GF}] = \lvert p\rvert^2$; equivariant master equation at $D = 4$ with the gauge-fixing vertices; negative controls
G2a–G2b	the classical breaking of Theorems 2.17, 2.18, its Bianchi identity, $\mathrm{LIM}\,\widehat{\Delta} = 0$
G3a–G3b	$\Delta_0 F = 0$; the regularised quantum action principle on two- and three-point functions, Route A against Route B
G4a–G4d	action counterterms, $\beta^{(1)} = 0$, the obstruction components and the restoring counterterms
G5	Konishi: $D_1 = 2\gamma_K$, $\gamma_K = \frac{3g^2N}{4\pi^2}$ (-48 units)
G6	$\mathbf{20'}$: $D_1 = 0$
G7	$\mathfrak{su}(2)$ and twist-two spectra
G8	relations (R2)–(R10) with and without restoration (Theorems 5.1, 5.3)
G9a–G9d	scheme D against scheme B; infrared safety; ξ_{GF}-independence; MS against MS-bar and two evanescent schemes
G10	finite parts from residues against full finite parts
G11	two loops: Konishi $\gamma^{(2)} = -\frac{3\lambda^2}{16\pi^4}$, $\mathbf{20'}$ $\gamma^{(2)} = 0$
G12	comparison with the heat-kernel track, rebuilt on a valid gauge fixing, through the invariants (I1)–(I6) of Definition 3.28 only
G13	joining and lifting on the Konishi extension and the closure $c_9 c_{10} = \frac{1}{2} c_1$ of [Bei03b, (3.36)], relation (R11)
G14	composite-operator breaking A_O on the $\mathbf{20'}$, stress-tensor and Konishi multiplets (Problem 7.2)
G15	two-loop special conformal constraint $[K^{(0)}, D^{(2)}] = [D^{(1)}, K^{(1)}]$ (Problem 7.5)

7 Testable questions on the supersymmetry of the schemes

In this formulation the supersymmetry of a dimensional scheme at a given loop order is the statement $\mathrm{Ob} = 0$ in the components along ξ, η and their arity-two closure, without finite counterterms; by Theorem 3.4 it is a property of the insertions of the evanescent breaking $\widehat{\Delta}$ alone [Bar98, §2.8]. High-loop computations of $\mathcal{N} = 4$ data in dimensional schemes [FSSZ08], [Vel08], [KLOV04] assume this

property at their loop order; no Slavnov–Taylor or supersymmetry Ward identity for composite operators has been evaluated in DRED, superspace DR or BMHV at any loop order [LIT, §2.3]. The questions below are those of [LIT, §5], numbered Q1–Q9, in the notation of this note; what the computation must reproduce is Definition 3.28.

Proposition 7.1 (No evanescent couplings in Scheme D)**.** The rotations of the internal space $E = Q(4-D)S \oplus W_{\mathbb{R}}^{\perp}$, acting on $A|_E$ as a vector and on λ through $\rho \circ q$, are an exact symmetry of S_D, Ψ and the non-minimal sector. Hence every counterterm of Scheme D is $\mathfrak{so}(E)$-covariant: the $4-D$ ϵ-scalars and the six scalars renormalise with one wave-function constant, one quartic and one Yukawa structure, all those of the reduction of ten-dimensional F_A^2 and $(\lambda, \not{D}_A\lambda)$; the separate evanescent couplings of dimensional reduction for non-supersymmetric gauge theories have no counterpart.

Proof. S_D is the reduction of the ten-dimensional action to QDS; rotations of the reduced directions commute with the derivatives along QDS and preserve the Hodge pairing, the Clifford relation and $(\rho(v)\zeta_1, \zeta_2) = (\zeta_1, \rho(v)\zeta_2)$; Ψ contains $A|_{QDS}$ only. The counterterms are local functionals built from the vertices and propagators, all $\mathfrak{so}(E)$-covariant, so the pole parts are $\mathfrak{so}(E)$-invariant; the invariant monomials of the relevant weight are those listed. □

Problem 7.2 (Q1: composite operators, Scheme D, one loop)**.** For a gauge-invariant O let A_O be the one-loop part of $L(O)$ along ξ in Scheme D with $\psi = 0$ and $F_1 = 0$ along ξ (Theorem 3.20: the Fierz vertex $\mathfrak{F}(\xi)$ on the cycles through two letters, the counterterm commutator $\mathcal{C}_\xi$, the operator breaking). Prerequisite: $\mathrm{Ob}^{(1)}(\xi) = 0$ and $\mathrm{Ob}^{(2)}(\xi, \xi') = 0$ at the level of the action (gate G4c; the ten-dimensional form of the counting rules [S05, (4.10)] decides it). Decide:

(a) $A_O = 0$ on the stress-tensor multiplet and the other half-BPS multiplets, in particular on the **20′** bilinears. By (I5) A_O is a coboundary $[\rho_0, e]$ there in every scheme; the question is whether Scheme D needs $e \neq 0$, i.e. a finite operator counterterm.

(b) On the Konishi multiplet, $A_O = [\rho_0, e]+$(the lifting): by (I4) and Remark 3.30 the lifting κ is non-zero and obeys $\kappa(\eta)\beta(\xi)c_0 = -\gamma_K(\xi, \eta)c_0$; it is not removable by finite counterterms. The question is the remainder: is it a coboundary, and with which e?

(c) $e = 0$ (no finite restoring operator counterterm, as for elementary fields at one loop in SQED and SQCD [HKS99], [HS01]) or $e \neq 0$. Letters $\bar{\lambda}\,\rho \cdots \rho\,\lambda$ lengthen the open ρ-strings through the operator and are the candidates for $e \neq 0$; $e \neq 0$ shifts the finite part of Z_O and two-loop mixing coefficients such as the Konishi-anomaly coefficient [BKRS01].

Loop order one; Route A with Route B as check; gates G4c, G14.

Problem 7.3 (Q2: Scheme B, one loop)**.** (a) $\mathrm{Ob} \neq 0$ already on elementary functions: compute the $\mathcal{N} = 4$ analogues of the one-loop SQED and SQCD restoring counterterms [HKS99], [HS01] (gauge–Yukawa and gauge–quartic relations) as the components of F_1^{B} (Theorem 3.12, gate G4d).

(b) The finite operator counterterms e^{B} for the **20′** and Konishi multiplets; before the R-restoration the $SO(4)_R$-singlet component of the **20′** and the Konishi bilinear share all exact quantum numbers and mix.

(c) Two loops: with F_1^{B} and e^{B} the two-loop dilatation operator equals that of Scheme D up to a similarity transformation (Theorem 3.22, [DG91], [HN95]); without them the **20′** acquires $\gamma^{(2)} \neq 0$.

(d) Is F_1^{B}, up to s-exact terms, the finite counterterm produced by the missing $4-D$ ϵ-scalars, LIM of $(4-D)\times$ the residues of the ϵ-scalar cycles of Scheme D? This identifies the one-loop conversion between the schemes as a canonical transformation (Proposition 3.27 (f)).

(e) With background $SU(4)_R$ gauge fields (position-dependent R-ghosts) the only non-exact class must be the 't Hooft anomaly, proportional to $(N^2-1)d^{abc}$; with constant ghosts it does not occur.

Loop order one for (a), (b), (d), (e); two for (c).

Problem 7.4 (Q3: superconformal closure on operators, one loop)**.** Relation (R11): on the $\mathfrak{su}(2|3)$ sector, c_9 and c_{10} are the coefficients of the length-changing parts of the classical supercharges [Bei03b, (3.32)], read from the tree-level BV data of Proposition 3.26 and Definition 2.12, c_1 is the normalisation of the one-loop dilatation operator from Theorem 3.22, and the order-g^2 closure of $\{Q, S\}$ requires $c_9 c_{10} = \frac{1}{2} c_1$ [Bei03b, (3.36)], the $\mathfrak{su}(2|3)$ form of $\kappa(\eta)\beta(\xi)c_0 = -\gamma(\xi,\eta)c_0$ [FR, Thm 4.9, §7]. Decide whether it holds in Scheme D with $F = 0$, and in Scheme B only with F_1^{B}, e^{B}. This turns the identification of the deformed $\mathfrak{su}(2|3)$ representation with renormalised operators, assumed in [Bei03b], into a computation. Loop order one; gate G13.

Problem 7.5 (Q4: conformal breaking at $D \neq 4$, one loop)**.** $\mathcal{N} = 4$ in D dimensions has no interacting fixed point; conformal symmetry is broken by $(4 - D) \int \sigma_v \mathscr{L}_0$ (Theorem 2.17 (ii), [DHKS07, (25), (27)]).

(a) Normalisation: Theorem 3.22, $D_1 = -2\,\mathrm{LIM}\,\mathcal{R}$, against the Konishi and $\mathbf{20'}$ values (gates G5, G6).

(b) K_1^0 of Theorem 3.24 is the one-loop special conformal anomaly; at two loops the order-$\hbar^2$ part of the (K, P)-component of $L \circ L = 0$, $[K^{(0)}, D^{(2)}] = [D^{(1)}, K^{(1)}]$, fixes the non-conformal part of the two-loop dilatation operator [BKM05].

(c) In Scheme B, does K_1 acquire the extra term $[\rho_0(K), e^{\mathrm{B}}]$ from the supersymmetry-restoring operator counterterm, i.e. do conformal covariance and supersymmetry restoration require the same finite counterterm?

Loop order one for (a), (c); two for the constraint in (b).

Problem 7.6 (Q5: the four-scalar identity, two loops)**.** The identity tying the quartic scalar vertex to g^2 and to the Yukawa vertex is established in dimensional reduction at one loop [S05, (4.8)] and, in a restricted limit, at three loops [LIT, §2.2]. In Scheme D: is the two-loop Fierz-insertion part of $\mathrm{Ob}^{(2)}$ on the four-scalar function zero? Proposition 7.1 fixes the pole structure (ϵ-scalars and scalars renormalise together); the question is the finite part. Loop order two.

Problem 7.7 (Q6: onset of the breaking of dimensional reduction, four and five loops)**.** Determine the ten-dimensional counting rule: the minimal length of a closed ρ-string in $Q4S \oplus W^{\perp}$ for which an insertion of $\mathfrak{F}$ survives in the formal algebra (the four-dimensional rule gives the five-index structure with trace $48D(D-1)(D-2)(D-3)(D-4)$ and four loops for the pure-SYM propagator [LIT, §2.4]). Test the claims that the insertion vanishes on the $\mathcal{N} = 4$ Yukawa vertex through four loops, with the first candidates at five loops (six flavour matrices in one trace), and settle the conflicting estimates of five versus eight or ten loops [LIT, §2.4]. Include the chirality-odd traces: by Remark 4.10 they enter no pole at one loop; find the first order at which a closed fermion loop with at least ten ρ's reaches a pole. Loop orders four and five, by counting and selected graphs.

Problem 7.8 (Q7: local coupling, one loop)**.** Promote the scale of the pairing of $\mathfrak{g}$ to a background function. The breakings $\widehat{\Delta}^{(1)}(\Delta)$, $\widehat{\Delta}^{(1)}(K_{v_1})$ are then the variations along σ_v-weighted coupling directions. Compute the one-loop local-coupling class of the constant-ghost Slavnov–Taylor identity in Scheme D (in the $\mathcal{N} = 1$ language it is $r_\eta = C(G)/8\pi^2$ for $\mathcal{N} = 1$ SYM [Kra02] and absent for $\mathcal{N} = 2$ [LIT, §3]); its vanishing for $\mathcal{N} = 4$ would make the Lagrangian-insertion formula $\partial_{g^2} \langle \cdots \rangle = \int \langle \mathscr{L} \cdots \rangle$ exact up to contact terms. Loop order one.

Problem 7.9 (Q8: deformed theories, one loop)**.** For β- and γ_i-deformed $\mathcal{N} = 4$ the local cochains must include double-trace couplings; their one-loop beta functions are the obstruction to the conformal components of the master equation; $U(N)$ and $SU(N)$ differ by the abelian factor. The construction of Sections 2–4 applies with the deformed vertices; not implemented in the engines of Section 6. Loop order one.

Problem 7.10 (Q9: γ_5 in the Standard Model)**.** The BV formulation of Scheme B with sterile partners (Proposition 2.5, the genuinely chiral case) is the setting of [BIMS20], [BIKMS21]. Statement to test: with symmetry-restoring counterterms the γ_5-odd four-loop coefficient of the strong beta function equals the reading-point value fixed by Weyl consistency conditions [LIT, §4]; smallest

test, a chiral gauge–Yukawa model at the order where two γ_5-odd traces first meet in a divergent graph. Outside the present engines; loop order four in the Standard Model.

Problem 7.11 (Classes across the tracks)**.** By Theorem 3.29 the data (I1)–(I6) agree between Scheme D, Scheme B and the heat-kernel track once the latter is built on a valid gauge fixing. A difference would show that (Hs) fails, i.e. an additional closed direction of the one-loop quantisation. Gate G12.

Remark 7.12 (What is settled where)**.** Q1, Q2 (a), (b), (d), Q3, Q4 (a), (c), Q7 and the class comparison are one-loop questions settled by Route A with Route B as the independent check; Q2 (c), Q4 (b), Q5 require the two-loop extension of Theorem 3.20 (three letters at most, [HK, DESIGN_STEP2, R8]) and of Lemma 3.10 [BIKMS21]; Q6 is a counting statement followed by selected graphs; Q8, Q9 apply the framework outside the undeformed $\mathcal{N} = 4$ theory.

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